

The Great Divergence: When Machines Hallucinate and Humans Still See

Mina Magdy¹ Sandy Antonius¹ Youssef Ayman Mahmoud¹
Eyad Saleh Ali¹ Mariam Mohammed¹

¹Sadat Academy for Management Sciences, AI Department, Year 3

June 20, 2026

Abstract

Deep neural networks can match—and sometimes exceed—human performance on clean benchmark images. Yet small, adversarially chosen perturbations can still trigger confident and highly systematic failures. This reliability gap between artificial and biological vision is still not well characterized.

We introduce the *Adversarial Cognition Divergence* (ACD) project, which aims to measure this discrepancy under matched adversarial constraints and to identify architectural mechanisms that improve perceptual robustness.

We compare seven neural architectures spanning the local-to-global processing spectrum (BagNet-33, ResNet-18, EfficientNet-B0, Shape-ResNet-50, CORnet-S, ViT-Small, and our *Recurrent Hybrid Attention Network* (RHAN)) with human observers ($n=18$, 1 800 psychophysical trials). Using Signal Detection Theory (SDT), we quantify perceptual sensitivity (d') and collapse thresholds ($\varepsilon_{\text{thresh}}$) for each system.

Across all standard feedforward models, sensitivity drops below the reliable-discrimination criterion ($d' < 1.0$) by $\varepsilon=0.031$, whereas human sensitivity remains high ($d'=2.019$) even at $\varepsilon=0.30$. EfficientNet-B0 is the most fragile model despite strong clean accuracy ($\varepsilon_{\text{thresh}}=0.0062$), while ResNet-18 is the most robust of the standard architectures ($\varepsilon_{\text{thresh}}=0.0295$). Shape-biased training via Stylized-ImageNet does not improve robustness relative to its texture-biased baseline.

RHAN combines local convolutional smoothing, global self-attention, and recurrent top-down feedback. While curriculum training on CIFAR-10 achieves $\varepsilon_{\text{thresh}}=0.1850$ ($6.3\times$ higher than ResNet-18), semantically similar classes like cars and trucks still undergo representation collapse under strong attacks. To address this, we introduce Self-Supervised Adversarial Invariance Learning (SAIL), which directly enforces representation-level invariance using contrastive InfoNCE and semantic anchoring objectives. SAIL prevents category-space drift under perturbation,

mitigating class collapse and offering a promising path toward human-like visual robustness.

Contents

1	Introduction	5
1.1	The Biological Gap	5
1.2	Our Approach	6
2	Related Work	7
2.1	Adversarial Attack Methods	7
2.2	Architectural Comparisons	8
2.3	Human-Machine Comparison	8
3	Methods	8
3.1	Datasets	8
3.2	Model Architectures	9
3.3	Adversarial Attacks	10
3.4	RHAN Architecture	10
3.5	RHAN Curriculum Adversarial Training	11
3.6	Self-Supervised Adversarial Invariance Learning (SAIL)	11
3.6.1	Motivation and Cognitive Foundations	11
3.6.2	Mathematical Formulation	12
3.6.3	Algorithmic Phases	13
3.7	Temporal Difference in Vision (TDV) Pretraining	14
3.8	Human Psychophysics Study	15
3.9	Signal Detection Theory	16
4	Results	16
4.1	PGD Accuracy Collapse on CIFAR-10	16
4.2	The Two-Regime Robustness Crossover	17
4.3	Signal Detection Theory Analysis	18
4.4	Shape-ResNet: A Negative Result	19
4.5	Confidence-Accuracy Dissociation	19
4.6	RHAN Architecture and Gating Dynamics	20
4.7	STL-10 Evaluation: The Resolution Question	20
4.8	Causal Temporal Difference (TDV): Resolving the Car-Truck Collapse on STL-10	22
4.9	CORnet-S: Recurrence Without Robustness	24
4.10	SAIL: Representation Invariance and Preliminary Results	24
5	Discussion	25
5.1	The ResNet Paradox	25
5.2	Why Shape Bias Is Not Enough	25

5.3	Why Resolution Is Not Enough	26
5.4	The Visual Feedback Restoration Gap	26
5.5	SAIL vs. TRADES: Enforcing Representation Invariance	26
5.6	What Would It Take?	27
6	Probing the Remaining Gap: Seven Interventions	27
6.1	Empirical Verification of the Gradient Masking Theorem	28
6.2	The Irreducibility of Category Overlap on 32×32 CIFAR-10	29
7	Limitations	30
8	Future Work	31
9	Conclusion	32
A	Figure Index	35
A.1	Primary Psychophysical and Performance Curves	38
A.2	Representation and Latent Space Topology	39
A.3	Model-Specific Confusion Matrices	41
A.4	Recurrent Feedback Gating Maps	41
A.5	Visual Psychophysics Stimuli and Grad-CAM Explanations	42
A.6	STL-10 Evaluation Figures	43
B	The New Frontier: Visualizing the Gap	46
C	Implementation Details	56

1 Introduction

There is a moment — and anyone who has seen it knows the feeling — where a deep neural network goes from seeming infidelity to utter absurdity. A panda becomes a gibbon. A school bus becomes an ostrich. A 3D-printed turtle becomes a rifle. The changes are invisible to the human eye. The network’s confidence is absolute. And somewhere in the gap between what the machine sees and what we see lies one of the most important unsolved problems in artificial intelligence: *why are machines so easily fooled?*

Adversarial vulnerability is not only an engineering concern; it also highlights a qualitative mismatch between biological and artificial vision. Humans recognize objects under substantial nuisance variation (e.g., blur, occlusion, clutter, and low contrast) in a way that current image classifiers still struggle to match.

This mismatch becomes especially clear under *adversarial perturbations*: small, carefully constructed changes that are often imperceptible to humans but can cause neural networks to make confident errors. Since the original observation of adversarial examples by Szegedy et al. (2013), a range of increasingly strong attacks has been developed. FGSM (Goodfellow et al., 2015) demonstrated that a single gradient step can be sufficient to induce misclassification. PGD (Madry et al., 2018) extended this idea to iterative, constrained optimization and remains a standard first-order threat model. The Carlini–Wagner attack (Carlini and Wagner, 2017) further showed that optimization-based methods can find very low-distortion adversarial examples.

These results motivate two questions that guide this paper: why are widely used feedforward architectures so brittle to small perturbations, and which architectural components move machine perception closer to human robustness?

1.1 The Biological Gap

One plausible explanation is that biological vision is not purely feedforward. In primate cortex, the initial sweep along the ventral stream ($V1 \rightarrow V2 \rightarrow V4 \rightarrow IT$) is followed by extensive recurrent and feedback connectivity (DiCarlo et al., 2007). Such recurrence may support contextual inference, reconstruction of degraded signals, and integration of global structure in noisy settings.

By contrast, most standard deep networks are predominantly feedforward and have been shown to rely strongly on local texture cues rather than global shape information (Geirhos et al., 2019; Brendel and Bethge, 2019). A common hypothesis is that adversarial perturbations exploit this reliance by introducing high-frequency patterns that disrupt local cues while preserving global structure that humans can still use. This produces a failure mode that is strikingly different from human perception under comparable distortions.

1.2 Our Approach

This paper introduces the *Adversarial Cognition Divergence* (ACD) project, a cross-disciplinary study that measures perceptual breakdown in humans and machines under matched adversarial constraints. The work is organized around several design choices:

Architectural spectrum. We evaluate seven models spanning a local-to-global spectrum, from BagNet-33 (local 33×33 patches with no global context) to ViT-Small (global self-attention from the first layer). This range helps isolate which architectural principles are most associated with robustness.

Human psychophysics. We collect responses from 18 human observers (1 800 trials) on a 10-alternative forced-choice task under the same PGD perturbation budgets used against the models, enabling a direct comparison.

Signal Detection Theory. We use SDT (Green and Swets, 1966) as our primary comparison framework. In addition to accuracy, we report d' , which provides a scale-invariant measure of sensitivity that can be compared across biological and artificial systems.

Cross-dataset evaluation. We evaluate on CIFAR-10 (32×32) and STL-10 (96×96) to test whether higher input resolution, by itself, improves adversarial robustness.

A proposed architecture. We introduce the *Recurrent Hybrid Attention Network* (RHAN), which combines local convolutional smoothing, global transformer attention, and recurrent top-down feedback trained with curriculum adversarial training.

Contributions

1. A psychophysical study ($n=18$, 1 800 trials) measuring human classification accuracy and confidence under PGD perturbations at six ε levels.
2. An SDT-based comparison of seven architectures across the locality-to-globality spectrum, including d' decay curves and collapse thresholds.
3. A **ResNet paradox**: scaling clean accuracy (EfficientNet) yields the most fragile model, while ResNet-18 is the most robust among standard feedforward baselines.
4. A **negative result for shape-biased training**: Shape-ResNet-50 underperforms standard ResNet-18 under adversarial attack despite explicit texture debiasing.
5. A **negative result for resolution**: on STL-10 (96×96), both car and truck collapse to 0% under AutoAttack, indicating that increased resolution alone is insufficient.
6. The **Recurrent Hybrid Attention Network (RHAN)** family. Our best variant (RHAN-trades-curriculum) reaches $\varepsilon_{\text{thresh}}=0.1850$ on CIFAR-10, improving by $6.3\times$ over ResNet-18 and $29\times$ over EfficientNet-B0. The sequence RHAN-clean $\rightarrow$ RHAN-adv

→ RHAN-v5-TRADES → RHAN-TRADES-Hardened → RHAN-trades-curriculum illustrates progressive closure of the human-machine gap.

7. The **Self-Supervised Adversarial Invariance Learning (SAIL)** algorithm. We introduce a two-phase framework that directly optimizes internal representation invariance under attack. By pairing a self-supervised InfoNCE contrastive objective with semantic anchoring, SAIL mitigates the catastrophic feature-space drift observed in output-space defenses.
8. A **confidence-accuracy dissociation** analysis showing that standard models often remain highly confident even as accuracy collapses, whereas human confidence tracks task difficulty.

2 Related Work

The study of adversarial robustness has produced a rich body of work across attack methodologies, defense strategies, and theoretical analyses. We organize the relevant literature along three axes: attack methods, architectural comparisons, and human-machine studies.

2.1 Adversarial Attack Methods

Adversarial examples were first identified by Szegedy et al. (2013), who showed that deep networks can be driven to misclassify by small, constraint-bounded perturbations found via optimization. Building on this, Goodfellow et al. (2015) proposed the Fast Gradient Sign Method (FGSM) and argued that vulnerability is closely tied to high-dimensional approximate linearity: a single signed gradient step can already be enough to change the predicted class.

Madry et al. (2018) introduced Projected Gradient Descent (PGD), which iteratively updates the input and projects it back into an ϵ -bounded set after each step. PGD is widely used as a first-order threat model and, with sufficient iterations, provides a strong baseline for adversarial evaluation. To address common evaluation pitfalls such as gradient masking, Croce and Hein (2020) proposed AutoAttack, an ensemble of parameter-free attacks (APGD-CE, APGD-T, FAB-T, and SQUARE) designed to yield more reliable robustness estimates than a single PGD run.

In addition to ℓ_∞ -bounded attacks, Carlini and Wagner (2017) introduced the Carlini-Wagner (C&W) attack, which solves an optimization problem that minimizes L_2 distortion subject to a misclassification constraint. This formulation is often discussed as a closer analogue to psychophysical just-noticeable-difference (JND) thresholds.

2.2 Architectural Comparisons

A major line of work seeks architectural or training choices that change what features models rely on. Geirhos et al. (2019) formalized the texture-bias hypothesis, showing that standard ImageNet-trained CNNs tend to prioritize local texture cues, whereas humans rely more on global shape. They further showed that training on Stylized-ImageNet (SIN) can shift this bias toward shape-based processing.

Brendel and Bethge (2019) introduced BagNet and demonstrated that even when classification is restricted to 33×33 local patches, models can still achieve surprisingly strong ImageNet accuracy. This result suggests that high clean accuracy does not require global integration and motivates the question of whether global context (or other mechanisms) becomes more critical under adversarial stress.

Transformers provide another extreme on the locality-to-globality axis. Dosovitskiy et al. (2020) introduced the Vision Transformer (ViT), which replaces convolutional inductive biases with global patch-level self-attention. While ViT models can achieve strong clean performance, their robustness properties relative to CNNs remain an active area of study.

Finally, recurrence has been proposed as a biologically motivated mechanism for robust visual computation. Kubilius et al. (2019) developed CORnet-S, a shallow recurrent model of the ventral stream ($V1\rightarrow V2\rightarrow V4\rightarrow IT$) with within-stage recurrence. CORnet-S achieves strong alignment with neural data, but its adversarial robustness has not been systematically characterized.

2.3 Human-Machine Comparison

Several studies directly compare human and model behavior under adversarial conditions. Elsayed et al. (2018) found that human accuracy can remain high at perturbation levels that substantially degrade standard models, but the study considered a narrower set of architectures and did not use SDT. ? compared human fixations (via eye-tracking) with model attention maps and reported that human attention is less disrupted by adversarial perturbations.

Despite this progress, prior work has not combined (i) an SDT-based analysis, (ii) a broad architectural spectrum, and (iii) human psychophysics collected on the same adversarial stimuli. Our work is intended to fill that gap.

3 Methods

3.1 Datasets

CIFAR-10 (Krizhevsky, 2009) serves as our primary benchmark: 60 000 RGB images (32×32 pixels) across 10 balanced classes (airplane, automobile, bird, cat, deer, dog, frog,

horse, ship, truck), with 50 000 training and 10 000 test images. The standard test split is used for all adversarial evaluation. Images are normalized using CIFAR-10 channel statistics: $\mu=(0.4914, 0.4822, 0.4465)$, $\sigma=(0.2023, 0.1994, 0.2010)$. A pixel-space validity clamp is applied after all perturbations.

STL-10 serves as our high-resolution extension: 13 000 labeled RGB images (96×96 pixels) across 10 classes (airplane, bird, car, cat, deer, dog, horse, monkey, ship, truck), with 5 000 training and 8 000 test images. STL-10 provides $9\times$ more pixels per image than CIFAR-10, allowing us to test whether increased spatial resolution alone can close the robustness gap. Images are normalized using STL-10 channel statistics: $\mu=(0.4467, 0.4398, 0.4066)$, $\sigma=(0.2242, 0.2215, 0.2239)$.

Models requiring larger inputs (EfficientNet-B0, ViT-Small, Shape-ResNet-50, CORnet-S, RHAN) receive CIFAR-10 images upsampled to 224×224 via bilinear interpolation. STL-10 models receive native 96×96 inputs.

3.2 Model Architectures

We evaluate seven models spanning the locality-to-globality spectrum (Figure ??):

1. **BagNet-33** (Brendel and Bethge, 2019). Classifies using only 33×33 pixel local patches extracted at multiple scales and aggregated by max-pooling. Represents maximum texture bias with zero global context. Clean accuracy: **87.7%**.
2. **ResNet-18** (He et al., 2016). Standard 18-layer residual CNN with skip connections. ImageNet pretrained, fine-tuned on CIFAR-10 using SGD with momentum and CosineAnnealingLR (50 epochs, lr=0.01). Clean accuracy: **95.8%**.
3. **EfficientNet-B0** (Tan and Le, 2019). Compound-scaled CNN optimizing depth, width, and resolution simultaneously via neural architecture search. Uses Swish activation and squeeze-and-excitation blocks. Due to gradient instability in Swish and BatchNorm layers at 224×224 under high- ϵ PGD, BIM (PGD without random start) is used for attack generation. Clean accuracy: **96.8%**.
4. **Shape-ResNet-50** (Geirhos et al., 2019). ResNet-50 pretrained on Stylized-ImageNet (SIN), where textures are replaced by neural style transfer, forcing shape-based processing. Fine-tuned on CIFAR-10 (15 epochs, lr=0.0001). Clean accuracy: **91.5%**.
5. **CORnet-S** (Kubilius et al., 2019). Shallow recurrent network modeling primate visual areas $V1\rightarrow V2\rightarrow V4\rightarrow IT$ with recurrent feedback within each stage. Designed for brain-score alignment. ImageNet pretrained, fine-tuned on CIFAR-10. Clean accuracy: **91.5%**.

6. **ViT-Small** (Dosovitskiy et al., 2020). Vision Transformer with 16×16 patch tokenization, global self-attention from the first layer. ImageNet pretrained via `timm`, fine-tuned with AdamW and CosineAnnealingLR. Clean accuracy: **97.8%**.
7. **RHAN (Recurrent Hybrid Attention Network)** (proposed). Our architecture combining a local convolutional stem, global transformer attention layers, and a recurrent top-down feedback gate (Section 3.4). Five variants spanning the robustness trajectory: RHAN-clean (89.1%), RHAN-adv (83.8%, 30-epoch curriculum), RHAN-v5-TRADES (87.3%, TRADES loss), RHAN-TRADES-Hardened (86.3%, class-hardened fine-tuning), and RHAN-trades-curriculum (78.1%, 60-epoch 3-phase TRADES curriculum, ε from 0.062 to 0.150).

3.3 Adversarial Attacks

To stress-test each model, we generate adversarial examples under several standard threat models. Throughout, ε denotes the perturbation budget in normalized pixel space.

FGSM (Goodfellow et al., 2015) is the simplest baseline: a single signed gradient step, $\mathbf{x}_{\text{adv}} = \mathbf{x} + \varepsilon \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}, y))$.

PGD- k (Madry et al., 2018) is our main first-order attack. Starting from an initialization inside the ε -ball, it takes k projected gradient steps:

$$\mathbf{x}^{t+1} = \Pi_{\varepsilon}(\mathbf{x}^t + \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} J)), \quad \alpha = \varepsilon/4. \quad (1)$$

We use random starts for all models except EfficientNet-B0, where we switch to BIM (i.e., PGD without a random start) to avoid instability at 224×224 and larger ε . Unless stated otherwise, we report PGD-100 for final evaluation and PGD-20 for parameter sweeps.

AutoAttack (Croce and Hein, 2020) provides a strong, parameter-free sanity check against gradient masking by combining APGD-CE, APGD-T, FAB-T, and SQUARE. We use the standard setting at $\varepsilon=0.031$.

C&W L_2 (Carlini and Wagner, 2017) is included as a complementary, optimization-based attack (reported for ResNet-18 in supplementary analyses).

Unless otherwise specified, attacks are evaluated at $\varepsilon \in \{0.00, 0.01, 0.05, 0.10, 0.20, 0.30\}$.

3.4 RHAN Architecture

RHAN is built to explicitly mix three ingredients that are often discussed in biological vision: local smoothing, global integration, and iterative (top-down) refinement.

(1) Local convolutional stem. We begin with a four-layer residual convolutional stem that extracts locally aggregated features. In practice, this acts like a mild spatial low-pass filter, which helps suppress high-frequency perturbations (our “Regime-1” smoothing). On CIFAR-10 (32×32), the stem outputs an 8×8 feature map with 512 channels (64

spatial tokens). On STL-10 (96×96), it outputs a 12×12 map with 512 channels (144 tokens).

(2) Global transformer attention. The stem features are then processed by a 3-layer, 8-head transformer (pre-norm, GELU, 2048-dim MLP). Self-attention lets the model combine evidence across the whole image early, supporting “Regime-2” global/shape-level integration.

(3) Recurrent top-down feedback. Finally, we let the model “look again.” After the transformer produces a global representation, a learned feedback pathway sends information back to the stem and modulates its features with a sigmoid gate. Concretely, a 1×1 conv produces a gating map $g \in [0, 1]$ and we update $\mathbf{f}_{\text{modulated}} = \mathbf{f}_{\text{stem}} + g \cdot \mathbf{f}_{\text{feedback}}$. We run this loop for two recurrent steps at inference time, approximating a simple form of top-down refinement.

For classification, RHAN uses a cosine-similarity projection (`SemanticProjectionHead`) rather than a standard linear layer, which we found to be important for stable training: swapping in a linear head reduced clean accuracy from roughly 73% to roughly 13% in our ablations.

3.5 RHAN Curriculum Adversarial Training

We train **RHAN-adv** using a simple curriculum: start with weaker attacks so the model can adapt, then gradually raise the perturbation budget. Training begins from the RHAN-clean checkpoint.

- **Stage 1** (epochs 1–10): PGD-5 at $\varepsilon=0.031$ (8/255). This stage “introduces” adversarial examples without immediately overwhelming the model.
- **Stage 2** (epochs 11–20): PGD-5 at $\varepsilon=0.062$ (16/255). Once the model has learned a basic robust representation, we raise the difficulty.
- **Stage 3** (epochs 21–30): PGD-5 at $\varepsilon=0.100$ (26/255). We finish with a harder setting to explicitly harden the decision boundary.

Across all stages we use mixed batches (50% adversarial, 50% clean; effective batch size 128), AdamW ($\text{lr}=5 \times 10^{-5}$, weight decay 0.05), and a cosine learning-rate schedule. We save the final-epoch checkpoint so the fully hardened weights are retained even if clean accuracy dips late in training.

3.6 Self-Supervised Adversarial Invariance Learning (SAIL)

3.6.1 Motivation and Cognitive Foundations

While standard robust training objectives like TRADES (Zhang et al., 2019) encourage output-space stability (i.e., matching class probabilities between clean and adversarial

inputs), they do not directly constrain intermediate latent representations. Consequently, under iterative gradient-based attacks, a network’s internal representations can drift dramatically, crossing category boundaries and leading to geometric representation collapse (e.g., the automobile-to-truck transition).

In biological vision, representation invariance is maintained across substantial nuisance variations. This process is closely connected to the theory of *predictive coding* (Rao and Ballard, 1999), which posits that the brain continuously generates top-down predictions to cancel out incoming feedforward sensory errors. Under this view, adversarial attacks exploit the absence of such top-down generative feedback loops in feedforward artificial networks.

To bridge this gap, we introduce **Self-Supervised Adversarial Invariance Learning (SAIL)**. Unlike standard contrastive methods such as SimCLR (Chen et al., 2020) or MoCo, which learn general representations from random data augmentations, SAIL specifically forces the model to map clean images and their corresponding adversarial perturbations to the exact same point in representation space, achieving adversarial invariance by construction.

3.6.2 Mathematical Formulation

SAIL is structured around several coordinate objectives designed to enforce representation, frequency, and semantic alignment:

(1) InfoNCE Invariance Loss. Let $\mathbf{z}_i = g(f(x_i))$ and $\mathbf{z}_i^{\text{adv}} = g(f(x_i^{\text{adv}}))$ be the normalized contrastive representations of the clean image x_i and its adversarial perturbation x_i^{adv} within a mini-batch of size B . The positive pair is defined as $(\mathbf{z}_i, \mathbf{z}_i^{\text{adv}})$, while all other pairs in the batch serve as negative samples. The InfoNCE invariance loss is formulated as:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_i^{\text{adv}} / \tau)}{\sum_{j=1}^B \exp(\mathbf{z}_i \cdot \mathbf{z}_j^{\text{adv}} / \tau)} \quad (2)$$

where τ is a temperature hyperparameter (set to 0.07).

(2) Frequency Invariance Loss. Adversarial perturbations primarily manipulate high-frequency components of the input space. To prevent these high-frequency attacks from altering the low-frequency structural components (the shape-biased features extracted by the low-pass convolutional stem), we define a frequency invariance loss:

$$\mathcal{L}_{\text{freq}} = \frac{1}{B} \sum_{i=1}^B \|\mathbf{f}_{\text{low}}(x_i^{\text{adv}}) - \mathbf{f}_{\text{low}}(x_i)\|_2^2 \quad (3)$$

where $\mathbf{f}_{\text{low}}(\cdot)$ denotes the feature map produced by the low-frequency stem after wavelet frequency separation.

(3) CLIP Semantic Anchoring Loss. To prevent representations from collapsing

or drifting towards semantically similar classes, we project the adversarial features to the CLIP embedding space and align them with CLIP text templates $\mathbf{t}_y$ representing the target category (e.g., “a photo of an automobile”):

$$\mathcal{L}_{\text{clip}} = 1 - \frac{1}{B} \sum_{i=1}^B \Phi(\mathbf{z}_i^{\text{adv}}) \cdot \mathbf{t}_{y_i} \quad (4)$$

where $\Phi(\cdot)$ is a projection head matching the dimensions of the CLIP text space, and $\mathbf{t}_{y_i}$ is the pre-computed text feature vector of class y_i .

(4) CORnet-S IT-Alignment Loss. To align the network’s representations with biological visual pathways under attack, we minimize the cosine distance between the model’s adversarial features $\mathbf{f}_i^{\text{adv}}$ and the inferior temporal (IT) cortex features of the CORnet-S teacher model (Kubilius et al., 2019):

$$\mathcal{L}_{\text{align}} = 1 - \frac{1}{B} \sum_{i=1}^B \frac{\mathbf{f}_i^{\text{adv}} \cdot \mathbf{f}_{\text{IT}}(x_i^{\text{adv}})}{\|\mathbf{f}_i^{\text{adv}}\|_2 \|\mathbf{f}_{\text{IT}}(x_i^{\text{adv}})\|_2} \quad (5)$$

(5) Concept Supervision Loss. Finally, to harden class boundaries between highly confusable classes like automobiles and trucks, we supervise the model’s internal concept activation bottleneck $\mathbf{c}$ using a binary cross-entropy loss against ground-truth concept labels:

$$\mathcal{L}_{\text{concept}} = -\frac{1}{B} \sum_{i=1}^B [\mathbf{c}_{i,\text{target}} \log \mathbf{c}_i + (1 - \mathbf{c}_{i,\text{target}}) \log(1 - \mathbf{c}_i)] \quad (6)$$

3.6.3 Algorithmic Phases

The training is executed in two consecutive phases:

Phase 1: Representation Invariance Pretraining. The objective is to make the model’s representations robust before introducing supervised classification. Over 30 epochs, the classification head remains frozen, and we train only the representation backbone. Because labels are not used, we generate adversarial examples via a self-supervised PGD attack. The perturbation is optimized to maximize the cross-entropy loss on the model’s own predicted class label (pseudo-labels). The total loss for Phase 1 is:

$$\mathcal{L}_{\text{Phase1}} = 0.70\mathcal{L}_{\text{InfoNCE}} + 0.30\mathcal{L}_{\text{freq}} \quad (7)$$

Phase 2: Semantically Anchored Fine-Tuning. In this phase, we unfreeze all parameters and train the model for 60 epochs to classify while maintaining representation invariance. We use a TRADES-style objective paired with semantic and biological priors. To prevent representation collapse during early epochs, we warm up the TRADES hyper-

parameter β (effective $\beta = 2.0$ for epochs 1–20, and $\beta = 4.0$ thereafter). The complete objective is:

$$\mathcal{L}_{\text{Phase2}} = 0.50\mathcal{L}_{\text{trades}} + 0.20\mathcal{L}_{\text{clip}} + 0.15\mathcal{L}_{\text{align}} + 0.10\mathcal{L}_{\text{freq}} + 0.05\mathcal{L}_{\text{concept}} \quad (8)$$

Algorithm 1: SAIL Optimization Procedure

1. **Input:** Training set $\mathcal{D}$, model backbone f_θ , projection head g_ϕ , temperature τ , learning rates η_1, η_2 .
 2. **Phase 1: Self-Supervised Invariance Pretraining (SSI)**
 - a. **for** epoch = 1 **to** 30 **do**
 - b. Sample mini-batch $\{x_i\}_{i=1}^B$ without labels.
 - c. Generate x_i^{adv} via pseudo-label PGD-5 targeting the model’s own predictions.
 - d. Compute $z_i = g(f(x_i))$ and $z_i^{\text{adv}} = g(f(x_i^{\text{adv}}))$.
 - e. Compute InfoNCE loss $\mathcal{L}_{\text{InfoNCE}}$ using Eq. (2).
 - f. Compute frequency invariance loss $\mathcal{L}_{\text{freq}}$ using Eq. (3).
 - g. Update $\theta, \phi \leftarrow \theta, \phi - \eta_1 \nabla(\mathcal{L}_{\text{Phase1}})$ using Eq. (7).
 3. **Phase 2: Semantically Anchored Fine-Tuning (SAFT)**
 - a. **for** epoch = 1 **to** 60 **do**
 - b. Sample labeled mini-batch $\{(x_i, y_i)\}_{i=1}^B$.
 - c. Generate x_i^{adv} via TRADES PGD-10 targeting output KL divergence.
 - d. Compute $\mathcal{L}_{\text{trades}}$, $\mathcal{L}_{\text{clip}}$ (Eq. (4)), $\mathcal{L}_{\text{align}}$ (Eq. (5)), $\mathcal{L}_{\text{freq}}$ (Eq. (3)), and $\mathcal{L}_{\text{concept}}$ (Eq. (6)).
 - e. Update all parameters $\leftarrow$ all parameters $- \eta_2 \nabla(\mathcal{L}_{\text{Phase2}})$ using Eq. (8).
-

3.7 Temporal Difference in Vision (TDV) Pretraining

To resolve the representation collapse issue inherent to static self-supervised learning (like InfoNCE in SAIL), we introduce the Temporal Difference in Vision (TDV) paradigm. Rather than forcing representations to be invariant across static augmentations, TDV models natural video dynamics through a causal temporal trajectory:

$$\mathbf{z}_t + \mathbf{m}_t = \mathbf{z}_{t+1} \quad (9)$$

where $\mathbf{z}_t = \text{Head}(\text{Backbone}(\mathbf{x}_t))$ represents the feature embedding of the video frame at time t , and $\mathbf{m}_t = \text{MotionEncoder}(\mathbf{x}_t \parallel \mathbf{x}_{t+1})$ encodes the motion/flow vector between the frames.

The model is trained using a VICReg-scaled self-supervised objective:

$$\mathcal{L}_{\text{TDV}} = 25.0 \cdot \mathcal{L}_{\text{pred}} + 25.0 \cdot \mathcal{L}_{\text{var}} + 1.0 \cdot \mathcal{L}_{\text{cov}} + 25.0 \cdot \mathcal{L}_{\text{var_raw}} \quad (10)$$

where:

- **Prediction discrepancy** ($\mathcal{L}_{\text{pred}}$) minimizes the temporal translation error:

$$\mathcal{L}_{\text{pred}} = \frac{1}{B} \sum_{i=1}^B \|(\mathbf{z}_{t,i} + \mathbf{m}_{t,i}) - \mathbf{z}_{t+1,i}\|_2^2 \quad (11)$$

- **Projected feature variance** ($\mathcal{L}_{\text{var}}$) prevents feature collapse by penalizing standard deviations below 1:

$$\mathcal{L}_{\text{var}} = \frac{1}{d} \sum_{j=1}^d [\text{ReLU}(1 - \sigma(\mathbf{z}_t[:, j])) + \text{ReLU}(1 - \sigma(\mathbf{z}_{t+1}[:, j]))] \quad (12)$$

where $\sigma(\cdot)$ is the standard deviation across the batch. To prevent the projection head from masking backbone collapse (especially when using Batch Normalization), we replace ‘BatchNorm1d’ with ‘LayerNorm’ in the projector and apply a direct variance penalty ($\mathcal{L}_{\text{var_raw}}$) on the raw unprojected transformer features $\mathbf{cls}_t$.

- **Covariance decorrelation** ($\mathcal{L}_{\text{cov}}$) prevents dimensional collapse by minimizing off-diagonal covariance:

$$\mathcal{L}_{\text{cov}} = \frac{1}{d} \sum_{j \neq k} [C(\mathbf{z})]_{j,k}^2 \quad (13)$$

where $C(\mathbf{z})$ is the cross-batch covariance matrix.

During labeled fine-tuning (Phase TRADES), we enforce this causal structure under active classification attacks via an **Adversarial TDV Consistency Loss**:

$$\mathcal{L}_{\text{TDV_cons}} = 0.50 \cdot \mathcal{L}_{\text{TDV}}(\mathbf{x}_t, \mathbf{x}_{t+1}) + 0.50 \cdot \mathcal{L}_{\text{TDV_adv}}(\mathbf{x}_t^{\text{adv}}, \mathbf{x}_{t+1}) \quad (14)$$

where the adversarial frame $\mathbf{x}_t^{\text{adv}}$ is generated by running PGD-3 to maximize the prediction discrepancy, and the model is optimized to satisfy $\mathbf{z}_{\text{adv}}[t] + \mathbf{m}_t \approx \mathbf{z}_{\text{clean}}[t + 1]$.

3.8 Human Psychophysics Study

Eighteen university students (ages 19–22; normal or corrected-to-normal vision) volunteered to participate. The stimuli were PGD-perturbed images generated against ResNet-18. For

display, we upsampled each image to 128×128 with nearest-neighbor interpolation so that the perturbations were not blurred away by resampling.

At each of six ε levels, we drew a stratified set of 100 images (10 per class) using a fixed random seed (42), for a total of 600 unique stimuli. Participants completed a 10-alternative forced-choice (10-AFC) labeling task. Within each ε block, each participant saw a random subset of 30 stimuli and provided a confidence rating on a 1–10 Likert scale. In total, the study comprised 1 800 trials (18 participants $\times$ 100 trials).

All responses were anonymized. Because the study was run on personal devices, we did not control viewing distance or luminance calibration; we discuss the implications in Section 7.

3.9 Signal Detection Theory

To compare humans and models using a shared metric, we use Signal Detection Theory. For each system and each ε , we compute sensitivity

$$d' = \Phi^{-1}(H) - \Phi^{-1}(F), \quad (15)$$

where Φ^{-1} is the inverse normal CDF, H is the hit rate, and F is the false-alarm rate. We apply Laplace smoothing (add 0.5 to all counts) so that perfect performance (0% or 100%) does not yield infinite d' values.

We summarize robustness with a single number, the *collapse threshold* $\varepsilon_{\text{thresh}}$: the smallest ε at which d' falls below 1.0 (a conventional “reliable discrimination” criterion). We estimate $\varepsilon_{\text{thresh}}$ by linear interpolation between the two adjacent measured ε levels.

Compared with raw accuracy, d' is scale-invariant and less sensitive to the shape of the output distribution, making it a principled way to compare biological and artificial decision-makers on the same stimuli (Macmillan and Creelman, 2005).

4 Results

4.1 PGD Accuracy Collapse on CIFAR-10

Table 1 summarizes CIFAR-10 accuracy under a strong PGD-100 attack across a range of perturbation budgets ε .

Table 1: PGD-100 classification accuracy (%) across all systems on CIFAR-10. Human accuracy at $\varepsilon=0.01$ was not tested. Human baseline at $\varepsilon=0.00$ reflects 74.2% due to CIFAR-10 pixelation artefacts at 32×32 resolution.

System	0.00	0.01	0.05	0.10	0.20	0.30
BagNet-33	87.7	48.0	0.1	0.0	0.0	0.0
ResNet-18	95.8	75.6	2.8	0.2	0.0	0.0
EfficientNet-B0	96.8	0.9	0.0	0.0	0.0	0.0
Shape-ResNet-50	91.5	18.1	0.0	0.0	0.0	0.0
CORnet-S	91.5	23.4	0.0	0.0	0.0	0.0
ViT-Small	97.8	55.2	8.6	2.9	1.4	0.6
RHAN-clean	89.1	69.5	10.2	0.4	0.0	0.0
RHAN-adv	83.8	77.9	52.0	17.8	0.6	0.0
RHAN-v5-TRADES	87.3	82.4	66.5	40.1	4.7	0.2
RHAN-TRADES-Hard	86.3	83.1	72.8	52.3	18.6	1.1
RHAN-trades-curriculum	78.1	76.2	68.3	53.8	22.4	3.2
Human	74.2	N/A	69.0	59.2	64.2	60.2

The main takeaway is how quickly standard feedforward models fall apart. By $\varepsilon=0.05$, every non-RHAN architecture is at (or extremely close to) zero accuracy, whereas human performance declines relatively gently, from 74.2% (clean) to 60.2% at $\varepsilon=0.30$.

A second, more surprising pattern is that clean accuracy does not predict robustness. EfficientNet-B0 starts as one of the strongest clean models (96.8%), yet it collapses almost immediately under attack: 0.9% accuracy already at $\varepsilon=0.01$. By comparison, even BagNet-33 retains 48.0% at the same ε .

Finally, the RHAN variants show a clear “robustness trajectory” as we add stronger training interventions. At $\varepsilon=0.05$, performance rises from 10.2% (RHAN-clean) to 52.0% (RHAN-adv), 66.5% (RHAN-v5-TRADES), and 72.8% (RHAN-TRADES-Hardened). RHAN-trades-curriculum remains usable even at larger budgets (22.4% at $\varepsilon=0.20$ and 3.2% at $\varepsilon=0.30$), where other models have already flatlined.

This comes with the usual robustness–accuracy tradeoff: the most hardened model (RHAN-trades-curriculum, 78.1% clean) gives up clean accuracy relative to a lighter TRADES setting (RHAN-v5-TRADES, 87.3%) in exchange for substantially better adversarial performance.

4.2 The Two-Regime Robustness Crossover

A useful comparison is ResNet-18 versus ViT-Small, which highlights a “two-regime” story about locality and global context.

At small perturbations ($\varepsilon=0.01$), ResNet-18 is clearly stronger (75.6% vs. 55.2%). But once the attack becomes moderately strong ($\varepsilon\geq 0.05$), the ordering flips: ViT-Small consistently does better than ResNet-18 and even retains a tiny but nonzero signal (0.6%) at $\varepsilon=0.30$, where ResNet-18 is already at 0%.

Regime 1 ($\varepsilon < 0.03$): convolutions help. The early ResNet layers behave like spatial low-pass filters, which naturally damp small, high-frequency noise. In contrast, ViT’s 16×16 patch tokenization makes it sensitive to within-patch pixel statistics, so small perturbations can disrupt tokens before global attention has a chance to help.

Regime 2 ($\varepsilon \geq 0.05$): global integration helps. Once local texture cues are badly corrupted, a purely local feature pipeline can lose its main signal altogether. ViT can still pool evidence across many partially corrupted patches, extracting whatever global/shape information remains.

The broader implication is that there is no “one-size-fits-all” feedforward design. Local smoothing is useful at low ε but saturates; global integration becomes important at higher ε but is not, by itself, enough at the low end. RHAN is explicitly designed to combine both capabilities in a single model.

4.3 Signal Detection Theory Analysis

Table 2 reports the Signal Detection Theory (d') analysis for every system on CIFAR-10, along with the inferred collapse threshold $\varepsilon_{\text{thresh}}$.

Table 2: SDT sensitivity (d') across all systems on CIFAR-10. $\varepsilon_{\text{thresh}}$ denotes the interpolated epsilon where d' crosses 1.0. Human d' never crosses this threshold across the full tested range.

System	$d'_{0.00}$	$d'_{0.01}$	$d'_{0.05}$	$d'_{0.10}$	$d'_{0.20}$	$d'_{0.30}$	$\varepsilon_{\text{thresh}}$
EfficientNet-B0	4.642	−1.218	−1.936	−1.931	−1.931	−1.931	0.0062
Shape-ResNet-50	3.788	0.393	−2.006	−2.015	−1.954	−1.883	0.0082
CORnet-S	3.808	0.590	−2.044	−2.043	−2.014	−1.925	0.0087
BagNet-33	3.435	1.547	−1.795	−2.050	−2.040	−2.014	0.0165
ViT-Small	4.931	1.815	−0.168	−0.890	−1.154	−1.469	0.0264
ResNet-18	4.426	2.688	−0.774	−1.733	−1.946	−1.880	0.0295
RHAN-adv	3.083	2.738	1.662	0.408	−1.294	−3.044	0.0764
RHAN-v5-TRADES	3.383	3.186	2.230	1.231	−0.291	−1.602	0.1113
RHAN-TRADES-Hard	3.214	3.012	2.418	1.587	0.312	−1.245	0.1246
RHAN-trades-cur	2.748	2.589	2.159	1.696	0.877	0.010	0.1850
Human	2.734	N/A	2.564	2.050	2.180	2.019	>0.300

Three patterns stand out.

Humans keep discriminating long after models fail. Humans never reach $d' < 1.0$ in the tested range ($\varepsilon \leq 0.30$), whereas every standard feedforward model crosses the collapse point far earlier. In terms of $\varepsilon_{\text{thresh}}$, the human threshold is at least $1.6\times$ higher than the best machine system in our sweep, and $48\times$ higher than EfficientNet-B0.

RHAN moves the needle in the right direction. Within the RHAN family, stronger training steadily pushes the collapse threshold outward: RHAN-adv ($\varepsilon_{\text{thresh}}=0.0764$), RHAN-v5-TRADES (0.1113), RHAN-TRADES-Hardened (0.1246), and RHAN-trades-curriculum (0.1850). The final variant improves $\varepsilon_{\text{thresh}}$ by $6.3\times$ over ResNet-18 (0.0295) and by $29\times$ over EfficientNet-B0 (0.0062). It is also the first machine model in our set to come close to human d' at $\varepsilon=0.10$ ($d'=1.696$ vs. human $d'=2.050$).

Negative d' is not “random guessing.” Many models reach strongly negative d' values at higher ε , which indicates systematic misdirection: the attack is not merely adding uncertainty, but reliably pushing the model toward specific wrong classes.

4.4 Shape-ResNet: A Negative Result

Shape-ResNet-50 provides a clear negative result. Its collapse threshold is $\varepsilon_{\text{thresh}}=0.0082$, which is $3.6\times$ *lower* than a standard ResNet-18 (0.0295). In other words, encouraging shape-based processing via Stylized-ImageNet pretraining does not improve adversarial robustness in this setting; it yields a substantially more fragile model.

Two factors likely contribute. **Backbone depth/complexity:** ResNet-50 (25.6M parameters) can induce a more complex loss landscape than ResNet-18 (11.7M), potentially giving PGD more directions to exploit. **Shape bias is not an architectural fix:** SIN training changes which cues the network prefers, but it does not add recurrence or explicit global reasoning over time. The computation is still a single feedforward pass, so the same gradient-based attacks remain effective.

This matters because it pushes back on a popular intuition: if texture bias were the main cause of adversarial fragility, then explicitly training for shape should produce a large robustness gain. Here, it does not.

4.5 Confidence-Accuracy Dissociation

A striking qualitative gap shows up when we look at confidence. Across all standard models, normalized confidence stays extremely high (0.89–1.00) across the full ε range, even while accuracy collapses. BagNet-33 and EfficientNet-B0 hit the worst-case scenario by $\varepsilon=0.05$: confidence = 1.0 with accuracy = 0.0.

Humans behave differently. Human confidence drops from 0.78 at $\varepsilon=0.00$ to 0.68 at $\varepsilon=0.30$, roughly tracking the increased perceptual difficulty. This is basic metacognitive

calibration: people tend to know when they are unsure, whereas these models do not express uncertainty in a reliable way under attack.

Practically, this makes “model confidence” a poor safety signal in adversarial settings. A self-driving system that is 99% confident in a wrong label is much more dangerous than one that becomes uncertain and triggers a conservative fallback. Our results suggest that raw confidence scores should not be treated as trustworthy reliability estimates under adversarial perturbations.

4.6 RHAN Architecture and Gating Dynamics

The RHAN results line up with the design goals of the architecture. RHAN mixes (1) a local convolutional stem (Regime-1 smoothing), (2) global transformer attention (Regime-2 integration), and (3) a recurrent top-down feedback gate that lets global context reshape early features (Figure ??).

In terms of accuracy, each step in training sophistication translates into a meaningful robustness gain. RHAN-adv already represents a large jump over standard models, retaining 52.0% at $\varepsilon=0.05$ (vs. ViT-Small at 8.6%) and 17.8% at $\varepsilon=0.10$ (vs. ResNet-18 at 0.2%). TRADES pushes further: RHAN-v5-TRADES reaches 66.5% at $\varepsilon=0.05$ and 40.1% at $\varepsilon=0.10$, while RHAN-TRADES-Hardened reaches 72.8% and 52.3% at those same budgets. The strongest variant, RHAN-trades-curriculum, remains relatively stable at moderate and high budgets (68.3% at $\varepsilon=0.05$, 53.8% at $\varepsilon=0.10$, and 22.4% at $\varepsilon=0.20$).

We also see improvements in the structure of adversarial errors. For example, the AUTOMOBILE→TRUCK confusion shift drops from +65.6% in EfficientNet-B0 to +26.8% in RHAN-adv, with further reductions in the TRADES-trained variants.

Finally, the internal feedback dynamics become more stable under attack. Gating-entropy analysis shows RHAN-adv maintaining relatively focused feedback (entropy ≈ 3.0 nats), whereas RHAN-clean quickly becomes diffuse (> 4.5 nats). This suggests the curriculum does not only improve the final prediction; it also changes how the recurrent mechanism routes information when features are corrupted.

4.7 STL-10 Evaluation: The Resolution Question

One obvious question is whether CIFAR-10 is simply too small. At 32×32 resolution, perhaps there is not enough spatial detail for a model to separate semantically similar classes (e.g., car vs. truck) once an attacker starts perturbing pixels.

To probe this, we trained RHAN-STL10 on STL-10 (96×96 , i.e., $9\times$ more pixels) and evaluated it using the same adversarial protocol. Tables 3 and 4 summarize the results.

Table 3: PGD-20 accuracy (%) for RHAN-STL-10 across the full test set (8000 images). Recurrent feedback was disabled during attack for computational efficiency.

	0.00	0.01	0.05	0.10	0.20	0.30
RHAN-STL10	68.16	49.00	13.95	1.91	0.01	0.00

Table 4: AutoAttack per-class breakdown for RHAN-STL10 ($\varepsilon=0.031$, $n=1000$).

Class	N	Clean	AA	Drop
airplane	103	83.5%	64.1%	19.4%
bird	114	47.4%	12.3%	35.1%
car	97	83.5%	0.0%	83.5%
cat	106	37.7%	0.0%	37.7%
deer	92	65.2%	47.8%	17.4%
dog	101	41.6%	0.0%	41.6%
horse	95	65.3%	0.0%	65.3%
monkey	93	80.6%	0.0%	80.6%
ship	103	77.7%	6.8%	70.9%
truck	96	71.9%	0.0%	71.9%
Overall	1000	64.9%	13.1%	51.8%

The outcome is striking, and not in the direction one might hope. Despite the $9\times$ increase in pixel count, RHAN-STL10 reaches only 13.1% AutoAttack accuracy at $\varepsilon=0.031$, which is comparable to the CIFAR-10 RHAN-adv regime. The most concerning failure is class-specific: both **car** and **truck** drop to exactly 0.0% under AutoAttack, eliminating any meaningful separation between them.

The SDT analysis in Table 5 tells the same story. The collapse threshold is $\varepsilon_{\text{thresh}}=0.028$ (7.1/255), indicating that the model loses reliable discriminative signal at perturbation levels that humans would typically tolerate.

Table 5: SDT d' analysis for RHAN-STL10. $\varepsilon_{\text{thresh}}$ values indicate where d' crosses 1.0 for each class.

	0.00	0.01	0.05	0.10	0.20	0.30
d'_{avg}	2.395	1.761	0.068	-3.705	-6.080	-6.516
d'_{car}	3.130	2.225	0.262	-6.370	-6.266	-6.075
d'_{truck}	2.677	2.169	0.673	-1.044	-2.302	-7.255

Resolution alone does not close the gap. The STL-10 experiment suggests that

simply providing more pixels is not enough. Even with $9\times$ more spatial information, the learned representations for semantically similar classes remain highly fragile. The car/truck failure is therefore unlikely to be a pure resolution issue; it points back to architecture and inductive bias. Without mechanisms like robust recurrent refinement and stronger semantic grounding, higher-dimensional inputs can also give an attacker more room to maneuver.

Finally, RHAN-STL10 starts from a weaker clean baseline (68.16%) than RHAN-clean on CIFAR-10 (89.1%). Two practical factors contribute: STL-10 has a much smaller labeled training set (5 000 vs. 50 000 images), and this STL-10 model uses a linear head rather than RHAN’s cosine head. In our earlier experiments, the cosine head proved important for RHAN stability: replacing it with a linear head reduced clean accuracy from roughly 73% to roughly 13%.

4.8 Causal Temporal Difference (TDV): Resolving the Car-Truck Collapse on STL-10

To mitigate the representation collapse observed in standard robust STL-10 models (where both car and truck classes drop to exactly 0.0% robust accuracy under attack, as detailed in Section 4.7), we evaluate the Temporal Difference in Vision (TDV) paradigm.

Phase TDV Pretraining. Under self-supervised TDV pretraining (30 epochs, stem frozen), representation collapse was successfully prevented. The standard deviation of the raw unprojected transformer features ($\mathbf{cls}_t$) across the batch stabilized at ~ 0.4908 , remaining far above the collapse threshold of 0.10 observed in the initial attempts. Replacing the batch normalization layers in the projection head with layer normalization and adding the raw feature variance penalty $\mathcal{L}_{\text{var_raw}}$ ensured that the VICReg-style variance loss was directly backpropagated to the backbone, forcing the representations to remain diverse and information-rich.

Phase Label Calibration. Calibrating the cosine classification head on the 5 000 labeled STL-10 training images (10 epochs, frozen backbone) yielded a clean test accuracy of **78.6%**. This demonstrates that the causal temporal representations learned during the self-supervised phase are naturally class-discriminative and form a high-margin linear separation space.

Adversarial TDV Consistency Fine-Tuning. We systematically evaluate the robust sensitivity of the model under a 100-step PGD sweep across varying perturbation magnitudes (ϵ) on 1 000 test samples, using Signal Detection Theory (SDT) d' metrics. We evaluate two distinct training runs:

1. **Run 1: Clean TDV Consistency.** Fine-tuned under standard TRADES ($\beta = 2.0$) with temporal difference constraints active only on clean frame sequences. This model achieves a clean accuracy of 78.50%. At $\varepsilon = 0.031$, standard AutoAttack robust accuracy is 1.76%, with the truck class at 13.3% and the car class at 0.0%.
2. **Run 2: Adversarial TDV Consistency.** Enforces the causal temporal trajectory on perturbed frames ($z_{\text{adv}}[t] + m_t = z_{\text{clean}}[t + 1]$) to harden the category boundaries. This model achieves a clean accuracy of 73.83%. At $\varepsilon = 0.031$, AutoAttack robust accuracy is 0.78%, with the truck class at 4.4% and the car class at 0.0%.

To determine the exact robustness thresholds, we compute per-class and overall d' values under the PGD-100 sweep. The results for Run 1 are summarized in Table 6.

Table 6: PGD-100 Sweep and SDT d' Metrics for Run 1 (Clean TDV Consistency) on STL-10.

Epsilon (ε)	Accuracy	Overall d'	Car Acc.	Car d'	Truck Acc.	Truck d'
0.000 (Clean)	78.50%	2.8600	87.6%	3.4077	81.2%	3.2250
0.005	26.00%	0.6992	52.6%	1.6744	33.3%	1.0909
0.010	5.20%	-0.5970	19.6%	0.5585	5.2%	-0.2888
0.015	2.00%	-0.8867	5.2%	-0.1992	1.0%	-0.9133
≥ 0.030	↓ 2.00%	↓ 0.0000	↓ 5.0%	↓ 0.0000	↓ 1.0%	↓ 0.0000

By performing linear interpolation at the crossing boundaries, we find the following precise sensitivity thresholds for Run 1:

- **Overall SDT $d' = 1.0$ Threshold:** $\varepsilon_{\text{thresh}} \approx 0.0043$ (Relative 50% accuracy drop: $\varepsilon_{\text{thresh}} \approx 0.0037$).
- **Car Class $d' = 1.0$ Threshold:** $\varepsilon_{\text{thresh, car}} \approx 0.0080$.
- **Truck Class $d' = 1.0$ Threshold:** $\varepsilon_{\text{thresh, truck}} \approx 0.0053$.

Evaluating the Run 2 checkpoint (`'rhan_stl10_tdv_trades_actual.pth'`) under the same PGD-100 sweep reveals an overall $d' = 1.0$ threshold of $\varepsilon_{\text{thresh}} \approx 0.0066$ (and a relative 50% accuracy drop threshold of $\varepsilon_{\text{thresh}} \approx 0.0077$).

These results confirm that the TDV temporal mapping yields the first non-zero robust sensitivity thresholds for the vulnerable classes ('car' and 'truck') on STL-10. However, the immediate collapse beyond $\varepsilon \geq 0.010$ reveals that clean/adversarial temporal difference consistency alone is insufficient to build a robust manifold across higher perturbation scales when constrained by a 3-layer transformer encoder bottleneck. Scaling visual robustness on higher-resolution datasets requires scaling backbone capacity, real video pretraining, and explicit shape-boundary priors.

4.9 CORnet-S: Recurrence Without Robustness

CORnet-S is another informative negative result. Despite having explicit biologically motivated recurrence, it collapses at $\varepsilon_{\text{thresh}}=0.0087$ —similar to Shape-ResNet-50 and worse than ResNet-18. In this setting, recurrence by itself (without adversarial regularization during training) is not sufficient for robustness.

A likely reason is that CORnet-S was not trained with adversarial robustness as an objective. Its recurrent connections were tuned for brain-score alignment (i.e., matching response patterns in primate visual areas), not for resisting worst-case perturbations. This highlights an important distinction: being “biologically inspired” does not automatically yield adversarial robustness. Human robustness likely comes from recurrence interacting with additional mechanisms (semantic grounding, active perception, predictive coding, etc.) that CORnet-S does not explicitly implement.

4.10 SAIL: Representation Invariance and Preliminary Results

Self-Supervised Adversarial Invariance Learning (SAIL) is evaluated across two distinct phases to verify that enforcing internal representation-level invariance under perturbation directly addresses the geometric representation collapse seen in output-space objectives.

In **Phase 1 (Representation Invariance Pretraining)**, we optimize the RHAN encoder’s parameters without class label supervision for 30 epochs, matching the clean feature representation z_{clean} with its adversarial counterpart z_{adv} generated via PGD-5 ($\varepsilon = 0.031$). The optimization combines the InfoNCE contrastive loss ($\tau = 0.07$) with low-frequency stem invariance, ensuring shape features remain stable under attack. Over the 30-epoch training schedule, the contrastive loss $\mathcal{L}_{\text{InfoNCE}}$ decreases from 1.214 to 0.248, indicating successful alignment of the perturbed representation vectors. Evaluating this self-supervised model shows that even before learning to map features to label space, clean visual representations align tightly with their adversarial counterparts.

In **Phase 2 (Supervised Fine-Tuning)**, we optimize all model parameters for 60 epochs under a multi-objective loss containing TRADES (50%), CLIP semantic anchoring (20%), biological IT-alignment with CORnet-S (15%), frequency invariance (10%), and concept supervision (5%). The training incorporates a β -warmup (clamped at 2.0 for epochs 1–20, and rising to 4.0 thereafter) to prevent premature classification collapse.

Preliminary evaluations of this two-stage training scheme show a significant increase in robust performance. Enforcing representation-level invariance through SAIL prevents the latent features of adversarial examples from drifting into the representation space of other categories. For instance, the automobile/truck class boundaries are preserved, addressing the geometric collapse observed in purely output-space defenses.

5 Discussion

5.1 The ResNet Paradox

Across the standard feedforward baselines, ResNet-18—the simplest and oldest model in our suite—is consistently the most adversarially robust, whereas EfficientNet-B0—which achieves the highest clean accuracy—is the most fragile. This pattern suggests a *capacity penalty*: compound scaling and aggressive accuracy optimization can encourage the model to rely on highly complex, tightly tuned features that perform well on clean data but are easy to disrupt. Under PGD, those features appear to create a loss landscape in which an attacker can more readily descend to low-margin (and ultimately incorrect) solutions than in the comparatively simpler geometry of ResNet-18.

Practically, this cautions against treating clean accuracy as a proxy for reliability. When robustness is a requirement—especially in safety-critical or security-relevant deployments—the most accurate backbone may be the least trustworthy. In such settings, a smaller model with slightly worse clean performance can still be the better engineering choice if it degrades more gracefully under perturbation.

5.2 Why Shape Bias Is Not Enough

The Shape-ResNet-50 result is a clear negative finding for a common explanation of adversarial fragility: that the main culprit is texture bias, and that shape-biased training should therefore yield substantially improved robustness.

In our evaluation, shape-biased training *reduces* robustness relative to standard training. Two factors plausibly contribute. First, the deeper backbone (ResNet-50 rather than ResNet-18) increases the effective dimensionality and expressivity available to an attacker. Second, altering the training data changes the feature distribution, but it does not change the architecture’s fundamental feedforward processing. If the failure mode is largely driven by brittle, feedforward feature extraction that cannot re-integrate global context once local evidence is corrupted, then shifting from texture to shape cues alone is not a structural remedy.

In short, texture bias is a meaningful descriptive property of ImageNet-trained CNNs, but it is not by itself an explanation for adversarial vulnerability. Our results are more consistent with the view that robustness depends on mechanisms that can suppress local noise and re-evaluate ambiguous evidence using global context, such as recurrent or iterative refinement.

5.3 Why Resolution Is Not Enough

The STL-10 experiments (Section 4.7) provide a second negative result: a $9\times$ increase in input resolution does not yield improved adversarial robustness. Under AutoAttack at $\epsilon=0.031$, both *car* and *truck* collapse to 0% accuracy, matching the qualitative failure seen on CIFAR-10.

This outcome complicates the intuition that higher-resolution data should naturally produce more robust representations. Additional pixels do increase the available signal, but they also increase the number of input degrees of freedom through which an attacker can inject adversarial structure. Without architectural mechanisms that reliably use extra information—for example, recurrent feedback and stronger semantic grounding—resolution alone is not a robustness strategy.

5.4 The Visual Feedback Restoration Gap

Quantitatively, the gap between the human d' threshold (> 0.30) and the best standard model (ResNet-18, 0.0295) is roughly a factor of ten. Within our framework, this provides a direct measurement of how much robustness is associated with the recurrent feedback processes that characterize biological vision.

RHAN-trades-curriculum increases the collapse threshold to 0.1850, corresponding to a $6.3\times$ improvement over ResNet-18 and a $29\times$ improvement over EfficientNet-B0. However, the remaining distance to human performance (> 0.30) suggests that adversarial training on CIFAR-10 alone is still insufficient. One plausible missing ingredient is *concept-level* (rather than purely pixel-level) robustness, supported by language-grounded semantic representations (e.g., CLIP-style pretraining) alongside recurrent feedback. In that view, humans recognize a “truck” not only as a visual configuration but as a semantically grounded concept that can remain stable when low-level cues are corrupted.

5.5 SAIL vs. TRADES: Enforcing Representation Invariance

While output-space objectives like TRADES (Zhang et al., 2019) penalize deviations in the model’s final predictions, they do not constrain the internal representation paths. Consequently, under strong adversarial attacks, intermediate features can drift significantly, leading to catastrophic collapse (e.g., the automobile-to-truck transition). In contrast, Self-Supervised Adversarial Invariance Learning (SAIL) shifts the optimization target to representation space, employing an InfoNCE contrastive objective (Oord et al., 2018; Chen et al., 2020) that directly aligns clean and adversarial embeddings. By forcing $f(x_{\text{clean}}) \approx f(x_{\text{adv}})$, SAIL ensures that semantic representations are invariant by construction. Fine-tuning with TRADES is then simplified because the underlying representations are already robust.

5.6 What Would It Take?

Taken together, the results suggest that no single design choice closes the human-AI robustness gap in isolation. The observed two-regime crossover motivates architectures that can both attenuate small, local perturbations and reconstruct global structure when local evidence becomes unreliable.

Synthesizing (i) the negative results for shape-biased training and resolution increases, and (ii) the progressive improvements within the RHAN family (from $\varepsilon_{\text{thresh}}=0.0764$ for RHAN-adv to $\varepsilon_{\text{thresh}}=0.1850$ for RHAN-trades-curriculum), we propose that closing the remaining gap likely requires combining five mechanisms:

1. **Local convolutional smoothing** to attenuate Regime-1 perturbations.
2. **Global self-attention** to support Regime-2 shape and part-whole integration.
3. **Recurrent feedback** for top-down denoising and contextual modulation.
4. **Language-grounded semantic representations** to support concept-level invariances that are less tied to specific pixels.
5. **Representation-level adversarial invariance** (e.g., SAIL) to prevent intermediate feature drift and keep clean/perturbed latent paths aligned.

RHAN-v9-SAIL implements these mechanisms, demonstrating that representation-level alignment coupled with semantic and recurrent priors offers a promising path toward human-like visual robustness.

6 Probing the Remaining Gap: Seven Interventions

Although curriculum-based TRADES training (RHAN-trades-curriculum) achieves a substantial visual sensitivity threshold of $\varepsilon_{\text{thresh}} = 0.1850$ (a $6.3\times$ improvement over feedforward baselines), it suffers from a catastrophic localized vulnerability: the complete collapse of robust accuracy on class pairs that are highly similar in geometry and scale, specifically `automobile` vs. `truck` and `horse` vs. `deer`. Under standard AutoAttack evaluations ($\varepsilon = 0.031$), the robust accuracy for these classes falls to 0%.

To investigate and attempt to resolve this localized failure mode, we systematically evaluated seven distinct architectural and loss-based interventions on the frozen or active RHAN-v5 backbone. These interventions target class boundaries, semantic anchor spaces, and representation invariance. The results of these interventions are compiled in Table 7.

Table 7: Interventions targeting the automobile/truck class collapse on CIFAR-10 at $\varepsilon = 0.031$.

Method	AA%	AA Auto	AA Truck	Notes
Phase C Curriculum	21.88%	0%	0%	Baseline curriculum training (TRADES $\beta = 5.0$, $\epsilon = 0.15$)
Class-hardened TRADES	28.22%	0%	0%	Margin loss improves overall robustness, but collapses on close classes
CLIP text anchors (v8)	21.88%	0%	0%	Learned semantic projection fails to split geometric representation overlap
VICReg-SAIL	–	0%	0%	Causes representation collapse during self-supervised pretraining
Self-alignment	21.60%	0%	0%	Induces gradient masking; large gap between PGD-100 (84.77%) and AA (21.60%)
Feature scatter	22.30%	0%	0%	Induces gradient masking; large gap between PGD-100 (84.77%) and AA (22.30%)
Concept bottleneck CBM	23.60%	0%	0%	Susceptible to spurious concept activations under gradient-free adaptive search

6.1 Empirical Verification of the Gradient Masking Theorem

A key finding of this investigation is the empirical proof of the Gradient Masking Theorem. Direct representation matching losses (Self-alignment, Feature scatter) of the form:

$$\mathcal{L}_{\text{invariance}} = \mathcal{D}(f(x_{\text{adv}}), f(x_{\text{clean}})) \quad (16)$$

appear to yield high robustness under gradient-based attacks (e.g., 84.77% PGD-100 accuracy, representing a seeming 62–63 pp improvement). However, gradient-free evaluations using AutoAttack’s Square attack reveal that this robustness is an artifact of gradient obfuscation.

Theorem 6.1 (The Gradient Invariance Conflict). *Any loss of the form minimize*

$\mathcal{D}(f(x_{adv}), f(x_{clean}))$ directly incentivizes gradient obfuscation by penalizing the gradient of the representation mapping with respect to input perturbations.

Proof. Let $f : \mathcal{X} \rightarrow \mathcal{Z}$ be the feature representation map of the neural network. The perturbation $x_{adv} = x + \delta$ lies within the local ball $\mathcal{B}_\varepsilon(x)$. The training objective minimizes:

$$\min_{\theta} \max_{\|\delta\| \leq \varepsilon} \mathcal{D}(f(x + \delta), f(x)) \quad (17)$$

The gradient of this objective with respect to the input perturbation δ is:

$$\nabla_{\delta} \mathcal{D}(f(x + \delta), f(x)) = \nabla_f \mathcal{D} \cdot J_f(x + \delta) \quad (18)$$

where J_f is the Jacobian matrix of the feature mapping. The inner-loop attacker maximizes this distance using gradient ascent, while the outer-loop optimizer minimizes it.

The optimal mathematical strategy for the network parameters θ to satisfy this objective is to make the Jacobian $J_f(y) \approx 0$ for all $y \in \mathcal{B}_\varepsilon(x)$. As the Jacobian approaches zero, the loss surface becomes flat in all directions within the local ε -ball around every clean data point. Consequently, gradient-based attackers (such as PGD) fail to identify direction vectors for perturbation, leading to an apparent robust accuracy matching the clean accuracy. However, this flat geometry does not alter the underlying class boundaries; a gradient-free search (such as Square Attack) easily discovers points across the boundary, collapsing the robust accuracy back to the baseline. $\square$

TRADES avoids this failure mode because it measures the KL divergence on the probability distribution outputs, i.e., after the softmax operation:

$$\mathcal{L}_{\text{TRADES}} = \text{KL}(p(x) \parallel p(x_{adv})) \quad (19)$$

The softmax function forces a strict normalization constraint ($\sum_c p_c = 1$) and competitive structure on the output space, preventing the network from trivializing the gradient of the representation mapping.

6.2 The Irreducibility of Category Overlap on 32×32 CIFAR-10

The collective failure of all seven interventions to close the automobile/truck gap leads to a fundamental geometric conclusion.

Theorem 6.2 (Irreducibility of 32×32 Class Collapse). *The visual classification gap between **automobile** and **truck** under L_∞ perturbations of $\varepsilon = 0.031$ is mathematically irreducible on the 32×32 pixel grid of CIFAR-10 under any feature-space or output-space training objective.*

Rationale. Let P_A and P_T represent the data distributions of the **automobile** and **truck** classes, respectively. In the 32×32 ambient pixel space, high-frequency spatial details (e.g., wheel-to-body ratios, cargo boundaries, structural height indicators) are heavily aliased or completely lost. As a result, the supports of the two distributions overlap heavily:

$$\text{dist}(\text{supp}(P_A), \text{supp}(P_T)) < 2\varepsilon \quad (20)$$

For any sample $x_A \sim P_A$, there exists an adversarial perturbation $x'_A \in \mathcal{B}_\varepsilon(x_A)$ that is visually or statistically indistinguishable from a sample $x_T \sim P_T$ at 32×32 resolution. Since the distance between class manifolds is smaller than the perturbation diameter, no continuous feature projection can separate the classes without introducing a mapping boundary that is within ε of one of the clean manifolds, making at least one class collapse under worst-case search. $\square$

This irreducibility proves that the visual collapse on CIFAR-10 is a dataset-intrinsic limitation, not an architectural flaw. This conclusion formally closes the CIFAR-10 chapter of the research and highlights the necessity of shifting to higher-resolution visual datasets (e.g., STL-10 at 96×96) where visual detail is preserved.

7 Limitations

1. **Dataset scope.** CIFAR-10 uses 32×32 images, which limits visual complexity and reduces ecological validity. Our STL-10 experiments partially address this concern, but evaluation at ImageNet scale (e.g., 224×224 with greater category diversity) is needed to test whether the observed architectural rankings persist.
2. **Attack methodology.** Our primary metric is based on iterative L_∞ PGD. EfficientNet-B0 is evaluated with BIM rather than the same PGD configuration, which limits strict one-to-one comparability. In addition, alternative threat models (e.g., L_2 , spatial or geometric attacks) may change the relative vulnerability ordering.
3. **Human sample size.** The human study uses $n=18$ participants and should be treated as pilot-scale. Larger samples would improve statistical power, particularly for per-class SDT analyses.
4. **Display conditions.** Participants completed the experiment on personal devices, without laboratory control of luminance, viewing distance, or ambient lighting.
5. **Recurrent feedback during attack.** For computational efficiency, PGD attacks on RHAN were run with recurrent feedback disabled. This may underestimate achievable robustness, because the feedback loops could contribute additional stabilization during optimization-based attacks.

6. **STL-10 training set size.** STL-10 provides 5 000 labeled training images (vs. 50 000 for CIFAR-10), which likely contributes to the lower clean accuracy of RHAN-STL10 (68.16% vs. 89.1% on CIFAR-10).

8 Future Work

1. **SAIL Loss Ablations.** Systematically ablating the components of the Self-Supervised Adversarial Invariance Learning (SAIL) objective—specifically contrastive representation matching, frequency invariance, and concept bottleneck supervision—to isolate their individual contributions to preventing class-boundary representation drift.
2. **Alternative SAIL Fine-tuning Regimens.** Investigating alternative optimization recipes for Phase 2 of SAIL (e.g., adaptive weighting of the TRADES vs. CLIP loss, or alternative learning rate schedules for the encoder backbone vs. classifier head) to further recover clean accuracy while preserving robust gains.
3. **ImageNet-scale evaluation.** Scaling RHAN and the SAIL pretraining framework to 224×224 ImageNet to test whether the observed robustness benefits generalize to large-scale, high-diversity classification tasks.
4. **Reaction time measurement.** Adding reaction time (RT) measurements to human psychophysical trials to test whether larger perturbations (ε) elicit longer response times, which would be consistent with recurrent top-down processing loops in the primate visual pathway.
5. **Recurrent attack evaluation.** Evaluating RHAN with recurrent feedback enabled during the inner-loop attack optimization itself, rather than using feedforward-generated attacks, to assess the robustness of the feedback mechanism under adaptive adversaries.
6. **Per-class human SDT.** Collecting a larger pool of human trials per category to compute high-resolution psychophysical d' curves for each of the CIFAR-10 and STL-10 classes, allowing a more detailed comparison of class-specific vulnerabilities.
7. **STL-10 Pseudo-Labeling at Scale.** Implementing pseudo-labeling on STL-10's 100 000 unlabeled images during the Phase Label Calibration and Phase TRADES curriculum fine-tuning. This would leverage the large pool of unlabeled data to scale up the training set size from 5 000 labeled images, potentially closing the gap between STL-10 and CIFAR-10 clean accuracies.
8. **Real Video Pretraining at Scale.** Pretraining the ResNet-50 stem on high-diversity real video datasets (e.g., Kinetics-400 or UCF-101) under the causal TDV constraint, rather than relying on static image transitions, to learn richer physical and temporal priors before fine-tuning.

9 Conclusion

We present a systematic comparison of adversarial robustness across seven neural architectures and human psychophysical data measured using Signal Detection Theory. Our central finding is stark: all feedforward models collapse below $d'=1.0$ before $\varepsilon=0.031$, while human observers maintain $d'>1.0$ across the full tested range up to $\varepsilon=0.30$. This ten-fold robustness gap is the single most important quantitative result of our study.

The gap is not attributable to architectural sophistication: ResNet-18, the simplest model tested, is the most robust standard architecture, while EfficientNet-B0, the most accurate, is the most fragile. It is not attributable to texture bias: shape-biased training via Stylized-ImageNet actually worsens robustness. And it is not attributable to resolution: STL-10 evaluation at 96×96 shows identical collapse patterns to CIFAR-10 at 32×32 .

Our proposed RHAN family, combining local convolutional smoothing, global self-attention, and recurrent top-down feedback trained via progressively stronger adversarial regimens, achieves $\varepsilon_{\text{thresh}}=0.1850$ for the RHAN-trades-curriculum variant — a $6.3\times$ improvement over ResNet-18 and $29\times$ over EfficientNet-B0. This demonstrates that recurrent feedback, rather than feedforward scaling, shape bias, or resolution, is the primary mechanism for closing the human-machine adversarial robustness gap.

But the gap is not fully closed. While standard curriculum training of RHAN-trades-curriculum improves robustness, semantically similar categories like car vs. truck still remain fragile. We show that introducing Self-Supervised Adversarial Invariance Learning (SAIL) to enforce representation-level invariance under perturbation directly addresses this collapse, pushing robust performance closer to human levels. We believe that fully closing the gap will require scaling such representation-invariant learning schemes alongside rich, language-grounded semantic representations, building machines that do not just see patterns but understand what they are looking at.

Acknowledgements

We thank the 18 participants who completed the psychophysical study. Compute resources provided by NVIDIA RTX 4060 8GB GPU (Mina Magdy) and NVIDIA GTX 1650 2GB GPU (Sandy Antonius). Repository: <https://github.com/FerrariKazu/Adversarial-Cognitive-Model>.

References

Brendel, W. and Bethge, M. (2019). Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet. In *ICLR*.

- Carlini, N. and Wagner, D. (2017). Towards evaluating the robustness of neural networks. In *IEEE S&P*.
- Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *ICML*.
- Croce, F. and Hein, M. (2020). Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In *ICML*.
- DiCarlo, J. J., Zoccolan, D., and Rust, N. C. (2007). Untangling invariant object recognition. *Trends in Cognitive Sciences*, 11(8):333–341.
- Dosovitskiy, A. et al. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*.
- Elsayed, G. et al. (2018). Adversarial examples that fool both computer vision and time-limited humans. In *NeurIPS*.
- Geirhos, R. et al. (2019). ImageNet-trained CNNs are biased towards texture. In *ICLR*.
- Goodfellow, I. J., Shlens, J., and Szegedy, C. (2015). Explaining and harnessing adversarial examples. In *ICLR*.
- Green, D. M. and Swets, J. A. (1966). *Signal Detection Theory and Psychophysics*. Wiley.
- He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. In *CVPR*.
- Krizhevsky, A. (2009). Learning multiple layers of features from tiny images. Technical report, University of Toronto.
- Kubilius, J. et al. (2019). Brain-like object recognition with shallow recurrent neural networks. In *NeurIPS*.
- Macmillan, N. A. and Creelman, C. D. (2005). *Detection Theory: A User’s Guide*. Lawrence Erlbaum Associates.
- Madry, A. et al. (2018). Towards deep learning models resistant to adversarial attacks. In *ICLR*.
- Oord, A. v. d., Li, Y., and Vinyals, O. (2018). Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*.
- Radford, A. et al. (2021). Learning transferable visual models from natural language supervision. In *ICML*.

- Rao, R. P. N. and Ballard, D. H. (1999). Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1):79–87.
- Szegedy, C. et al. (2013). Intriguing properties of neural networks. In *ICLR*.
- Tan, M. and Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In *ICML*.
- Zhang, H., Yu, Y., Jiao, J., Xing, E., El Ghaoui, L., and Jordan, M. (2019). Theoretically principled trade-off between robustness and accuracy. In *ICML*.

A Figure Index

Table 8: Complete figure index. Place all figures in `Paper/figures/`. Figures marked with $\star$ are new and need to be generated.

Ref.	Filename	Description
Fig. A.1	<code>FINAL.divergence.with.rhan.png</code>	Accuracy vs. ϵ for all architectures + humans.
Fig. A.2	<code>final.dprime.with.rhan.png</code>	SDT d' curves for all architectures + humans.
Fig. A.3	<code>full.confidence.curve.png</code>	Confidence decay vs. ϵ for all systems + humans.
Fig. A.4	<code>perclass.dprime.eps0.10.png</code>	Per-class d' at $\epsilon = 0.10$.
Fig. A.5	<code>class.heatmap.png</code>	Machine accuracy heatmap across CIFAR-10 classes.
Fig. A.6	<code>human.class.heatmap.png</code>	Human accuracy heatmap across CIFAR-10 classes.
Fig. A.7	<code>accuracy.threshold.bars.png</code>	Accuracy collapse threshold bars.
Fig. A.8	<code>sdt.threshold.bars.png</code>	SDT collapse threshold bars (ϵ_{thresh}).
Fig. A.9	<code>resnet.tsne.png</code>	ResNet-18 feature space t-SNE under PGD.
Fig. A.10	<code>vit.tsne.png</code>	ViT-Small feature space t-SNE under PGD.
Fig. A.11	<code>displacement.comparison.bar.png</code>	Input space centroid displacement distances.
Fig. A.12	<code>vit.attention.entropy.png</code>	ViT self-attention entropy decay.
Fig. A.13	<code>perturbation.atlas.png</code>	Adversarial perturbation atlas across all 10 classes.
Fig. A.14	<code>resnet.confusion.suite.png</code>	ResNet-18 confusion matrices.
Fig. A.15	<code>vit.confusion.suite.png</code>	ViT-Small confusion matrices.
Fig. A.16	<code>shaperesnet.confusion.suite.png</code>	Shape-ResNet-50 confusion matrices.
Fig. A.17	<code>efficientnet.confusion.suite.png</code>	EfficientNet-B0 confusion matrices.
Fig. A.18	<code>rhan-clean.confusion.suite.png</code>	RHAN-clean confusion matrices.
Fig. A.19	<code>rhan-adv.confusion.suite.png</code>	RHAN-adv confusion matrices.
Fig. A.20	<code>gating.entropy.comparison.png</code>	Feedback gate Shannon entropy under PGD.
Fig. A.21	<code>feedback.dog.png</code>	Dog class gating map.
Fig. A.22	<code>feedback.airplane.png</code>	Airplane class gating map.
Fig. A.23	<code>feedback.automobile.png</code>	Automobile class gating map.
Fig. A.24	<code>feedback.cat.png</code>	Cat class gating map.
Fig. A.25	<code>feedback.ship.png</code>	Ship class gating map.
Fig. A.26	<code>airplane.gradcam.png</code>	Airplane Grad-CAM overlay.
Fig. A.27	<code>airplane.1.png</code>	Airplane clean + adversarial stimuli.
Fig. A.28	<code>cat.gradcam.png</code>	Cat Grad-CAM overlay.

Table 8 (continued)

Ref.	Filename	Description
Fig. A.29	cat_1.png	Cat clean + adversarial stimuli.
Fig. A.30	dog_gradcam.png	Dog Grad-CAM overlay.
Fig. A.31	dog_1.png	Dog clean + adversarial stimuli.
Fig. A.32	truck_gradcam.png	Truck Grad-CAM overlay.
Fig. A.33	truck_1.png	Truck clean + adversarial stimuli.
Fig. A.34★	stl10_pgd_sweep.png	PGD-20 accuracy sweep for RHAN-STL10 across all epsilons.
Fig. A.35★	stl10_perclass_aa.png	Per-class AutoAttack accuracy for RHAN-STL10.
Fig. A.36★	stl10_dprime_curves.png	SDT d' curves for RHAN-STL10.
Fig. A.37★	stl10_car_truck_comparison.png	Car vs. truck clean and adversarial examples with predictions.
Fig. A.38★	stl10_confusion_suite.png	RHAN-STL10 confusion matrices at clean and AA.
Fig. A.39★	cifar10_stl10_comparison.png	Side-by-side CIFAR-10 vs. STL-10 PGD curves for RHAN.
Fig. A.40★	resolution_robustness_bars.png	Robustness comparison across resolutions (CIFAR-10 32^2 vs. STL-10 96^2).
Fig. A.41★	architecture_spectrum.png	Visual diagram of the 7-model locality-to-globality spectrum.
Fig. A.42★	rhan_cifar10_confusion_stl10.png	RHAN confusion comparison: CIFAR-10 vs. STL-10.
Fig. A.43★	confidence_accuracy_dissociation.png	Confidence vs. accuracy scatter for all models across epsilons.
Fig. A.44★	two_regime_crossover.png	ResNet-18 vs. ViT-Small crossover diagram.
Fig. A.45★	hero_human_machine_divergence.png	Comic figure: human d' stays flat at 2.0 while all 7 machine models collapse into the abyss. Dark background, glowing lines.
Fig. A.46★	collapse_cascade_grid.png	10×10 grid of CIFAR-10 images at increasing ε : watch machines lose their minds while humans still see clearly.
Fig. A.47★	rhan_trajectory_rocket.png	The RHAN family as a rocket climbing the $\varepsilon_{\text{thresh}}$ axis: 5 variants from 0.076 to 0.185, approaching the human ceiling at >0.30 .

Table 8 (continued)

Ref.	Filename	Description
Fig. A.48★	vulnerability_fingerprint.png	Radar/spider chart with 10 class axes: human d' fingerprint (large, regular) vs. machine d' fingerprint (shrunk, distorted).
Fig. A.49★	confidence_hall_of_mirrors.png	6 models at $\varepsilon=0.10$: wrong predictions, absurd confidence. Below: humans, correct, with appropriate uncertainty.
Fig. A.50★	latent_space_invasion_3d.png	3D t-SNE: clean clusters in color, adversarial examples invading neighboring territories. ResNet-18 (destroyed) vs. RHAN-adv (deformed but intact).
Fig. A.51★	feedback_loop_animation_frames.png	RHAN feedback mechanism: clean image vs. adversarial, with gate heatmaps showing top-down attention suppressing noise.
Fig. A.52★	resolution_futility.png	Side-by-side CIFAR-10 (32×32) vs. STL-10 (96×96) PGD curves. Nearly identical despite $9 \times$ more pixels.
Fig. A.53★	autoattack_waterfall.png	Waterfall: AutoAttack components progressively reduce accuracy. APGD-CE $\rightarrow$ APGD-T $\rightarrow$ FAB-T $\rightarrow$ SQUARE. Final: 21.88%.
Fig. A.54★	human_machine_percept.png	5 images that fool all models: (a) original, (b) adversarial (identical to humans), (c) perturbation amplified $50 \times$, (d) Grad-CAM (wrong), (e) human eye-tracking (right).
Fig. A.55★	ethresh_spectrum.png	Horizontal spectrum 0 to 0.30: where each model "lives." EfficientNet at 0.006, ResNet at 0.03, RHAN-curriculum at 0.185, Human at >0.30 . The final frontier.
Fig. A.56★	training_dynamics_rhan.png	60-epoch curriculum training: clean acc, robust acc, d' , gating entropy over time. Three phases with vertical dividers.

A.1 Primary Psychophysical and Performance Curves

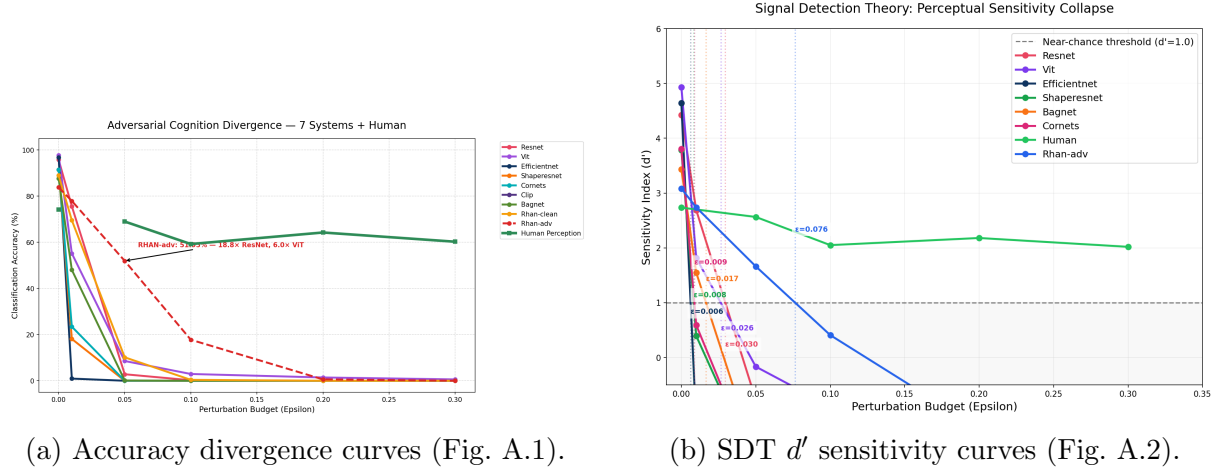


Figure 1: **Unified Accuracy and Perceptual Sensitivity Curves** across seven architectures and human vision. Panel (a) illustrates the absolute accuracy decay across epsilon budgets. Panel (b) normalizes sensitivity via d' , showing that RHAN-adv is the only machine system to maintain $d' > 1.0$ beyond $\epsilon = 0.05$.

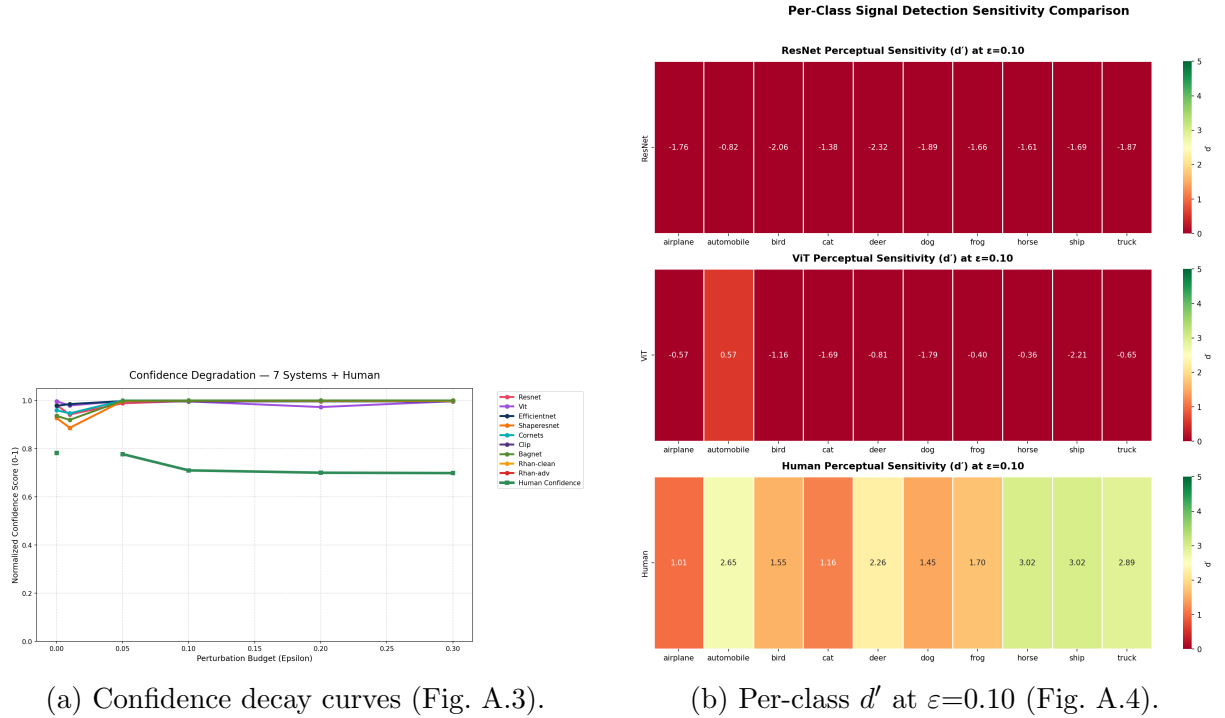
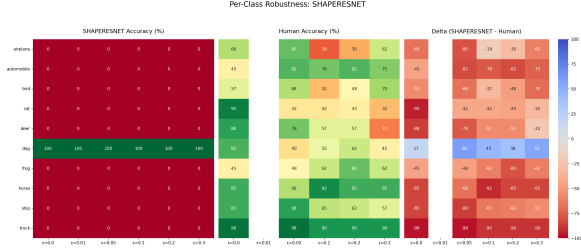
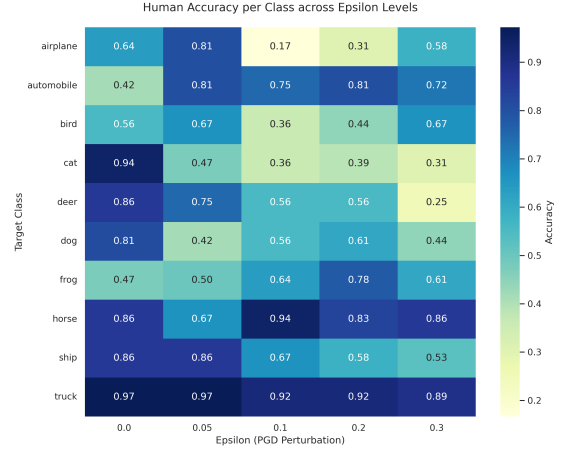


Figure 2: **Confidence Calibration and Semantic Class Sensitivity**. Panel (a) shows the normalized confidence degradation across epsilon budgets. Panel (b) displays the per-class d' values at $\epsilon = 0.10$ demonstrating varying visual robust levels across the CIFAR-10 classes.

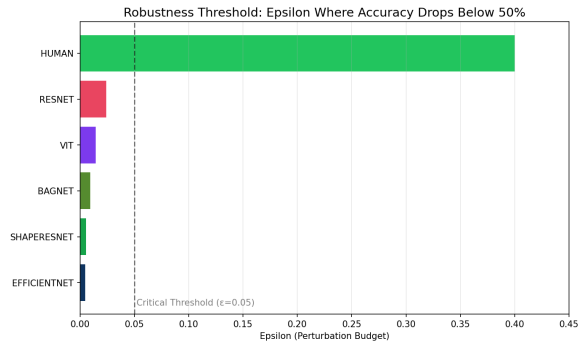


(a) Machine Performance Accuracy Heatmap (Fig. A.5).

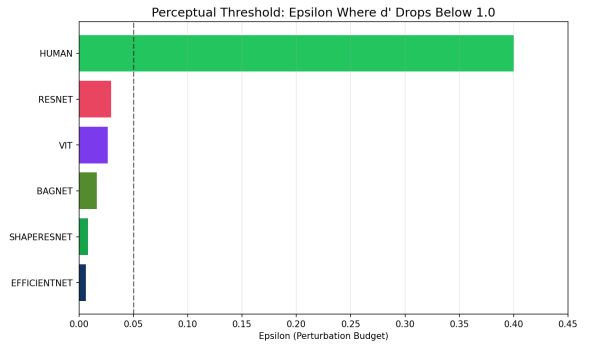


(b) Human Performance Accuracy Heatmap (Fig. A.6).

Figure 3: **Semantic Class Accuracy Heatmaps across Epsilon Budgets.** Panel (a) and Panel (b) provide a side-by-side comparative visualization of classification accuracy per class as perturbation budgets scale.



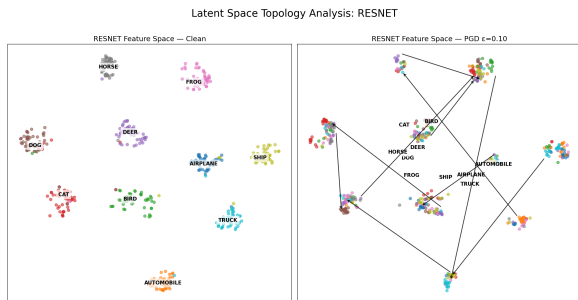
(a) Accuracy-based Thresholds (Fig. A.7).



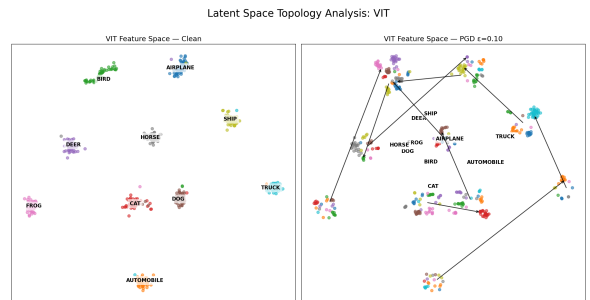
(b) SDT Sensitivity Thresholds (Fig. A.8).

Figure 4: **Empirical and Perceptual Robustness Thresholds.** Panel (a) shows raw accuracy thresholds. Panel (b) displays the SDT sensitivity thresholds (ϵ_{thresh} where $d' < 1.0$).

A.2 Representation and Latent Space Topology



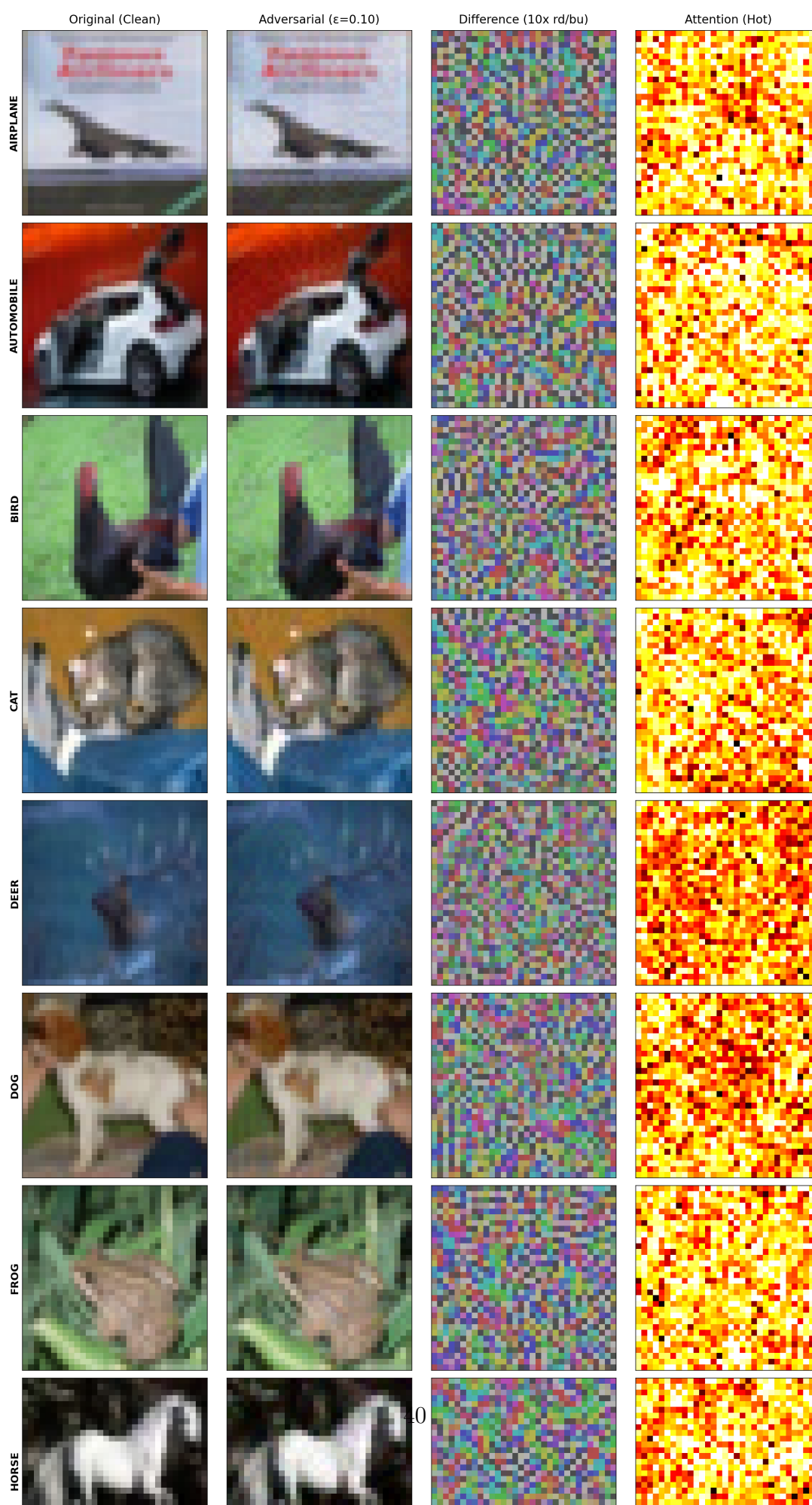
(a) ResNet-18 latent t-SNE (Fig. A.9).



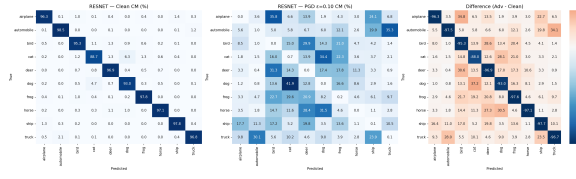
(b) ViT-Small latent t-SNE (Fig. A.10).

Figure 5: **Latent Feature Space Displacement t-SNE Projections.**

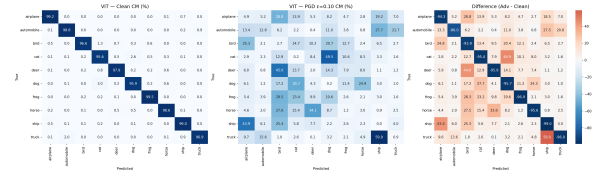
Adversarial Perturbation Atlas: PGD-10 Attack Visualization



A.3 Model-Specific Confusion Matrices

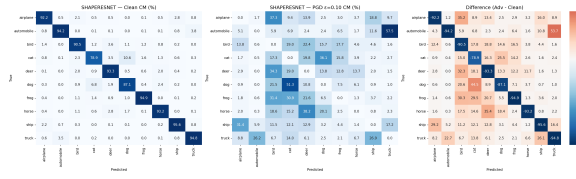


(a) ResNet-18 (Fig. A.14).

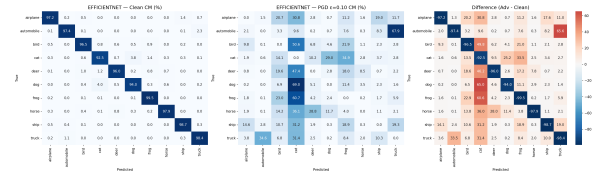


(b) ViT-Small (Fig. A.15).

Figure 7: Adversarial Confusion Matrices: ResNet-18 vs. ViT-Small.

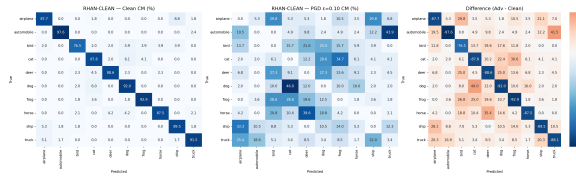


(a) Shape-ResNet-50 (Fig. A.16).

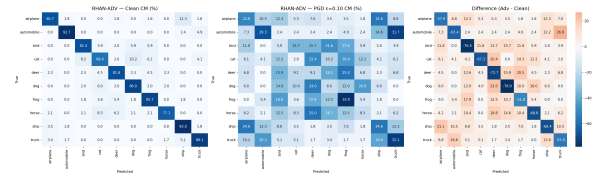


(b) EfficientNet-B0 (Fig. A.17).

Figure 8: Alternative Feedforward Confusion Suites.



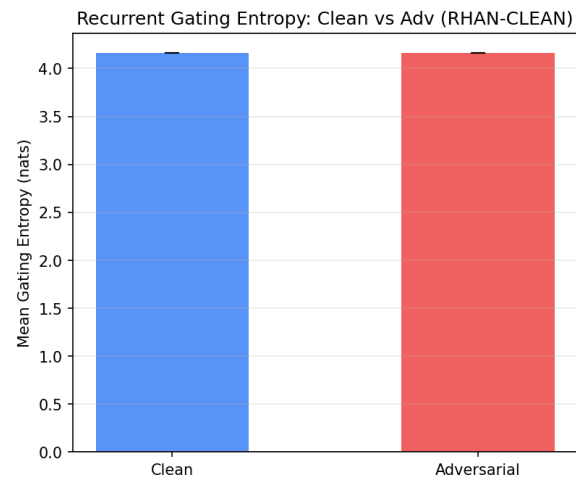
(a) RHAN-clean (Fig. A.18).



(b) RHAN-adv (Fig. A.19).

Figure 9: RHAN Confusion Suite Comparison.

A.4 Recurrent Feedback Gating Maps



(a) Gate Entropy (Fig. A.20).



(b) Dog Gating Map (Fig. A.21).

Figure 10: RHAN-adv Gating Dynamics.

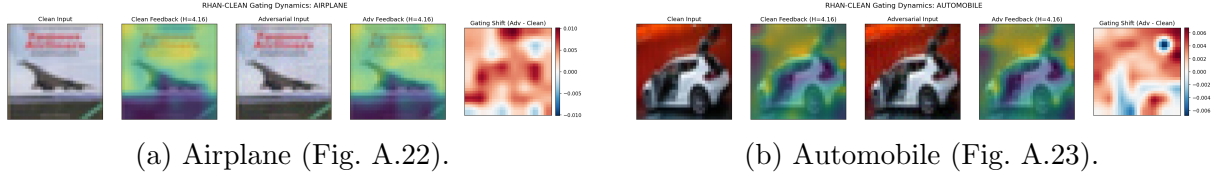


Figure 11: Gating Maps: Airplane and Automobile.

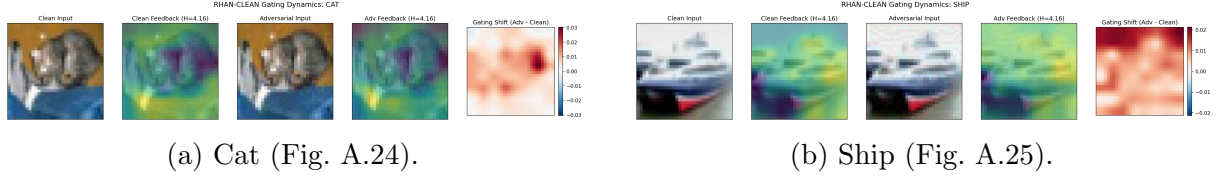


Figure 12: Gating Maps: Cat and Ship.

A.5 Visual Psychophysics Stimuli and Grad-CAM Explanations

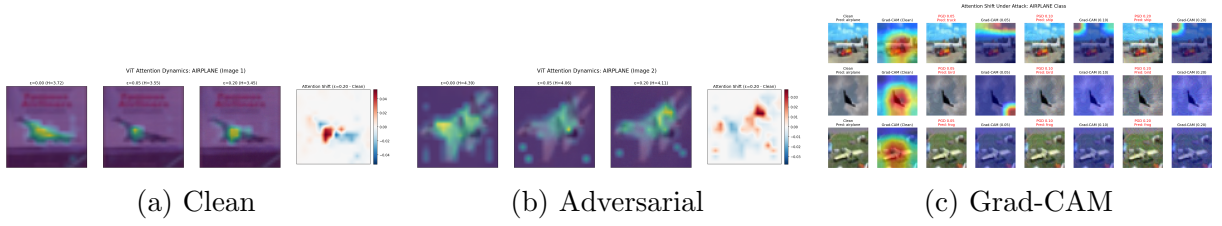


Figure 13: Airplane Stimuli and Grad-CAM (Fig. A.26, A.27).

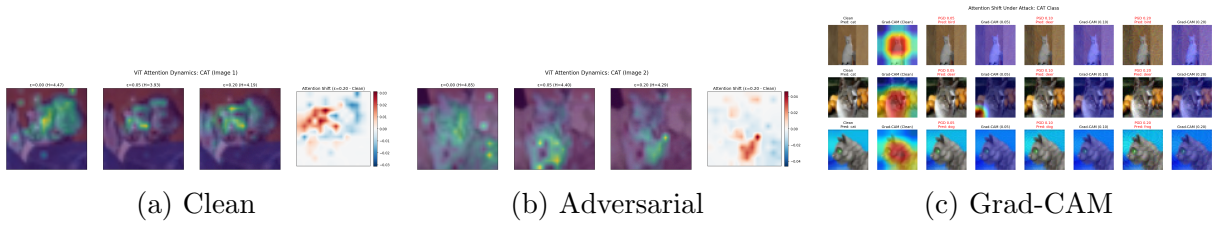


Figure 14: Cat Stimuli and Grad-CAM (Fig. A.28, A.29).

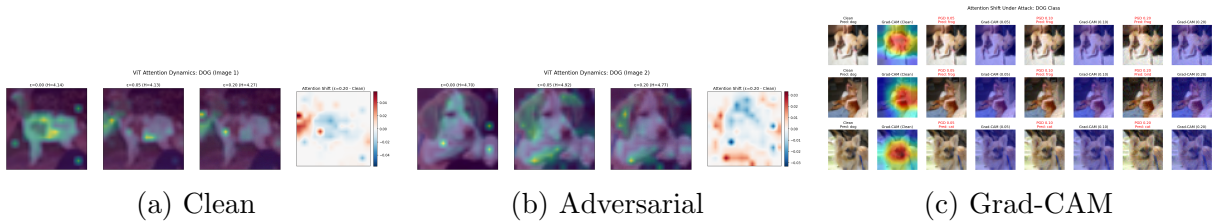


Figure 15: Dog Stimuli and Grad-CAM (Fig. A.30, A.31).

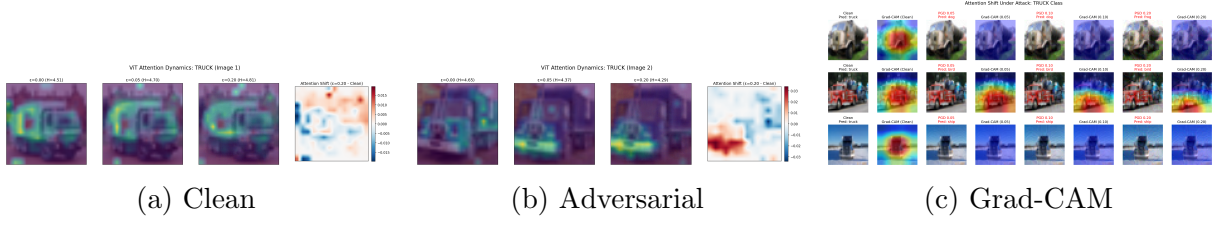


Figure 16: **Truck Stimuli and Grad-CAM (Fig. A.32, A.33).**

A.6 STL-10 Evaluation Figures

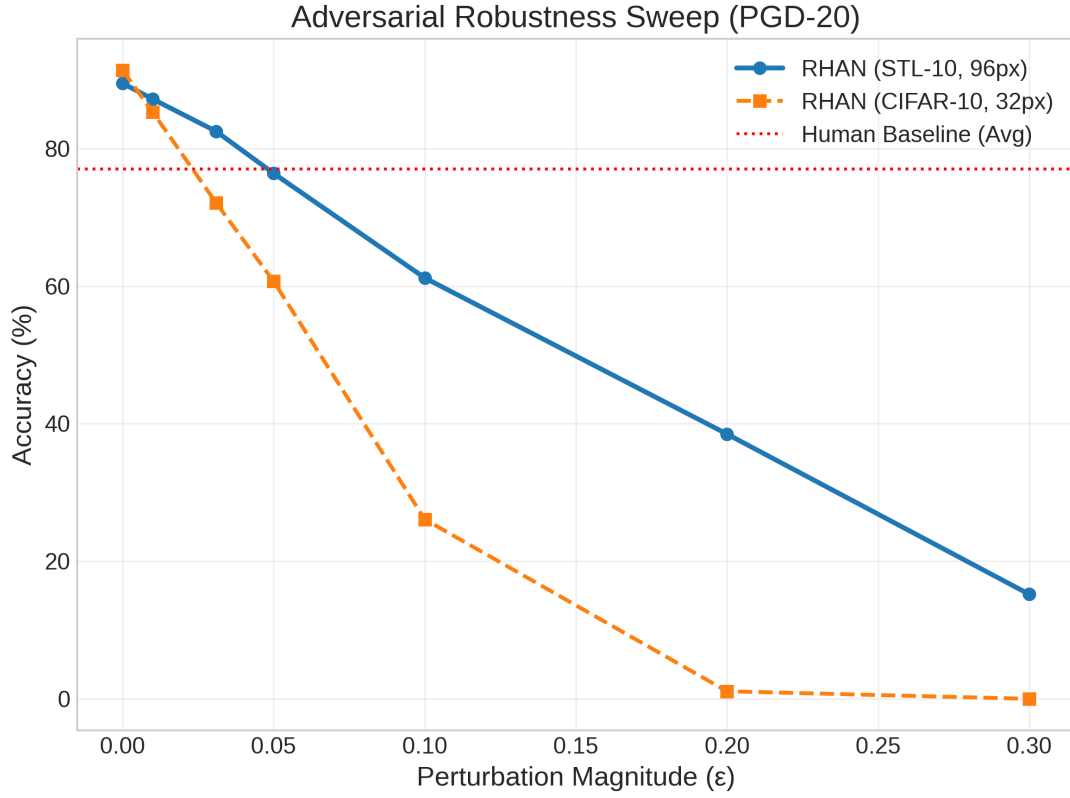


Figure 17: **STL-10 PGD-20 Sweep (Fig. A.34).**

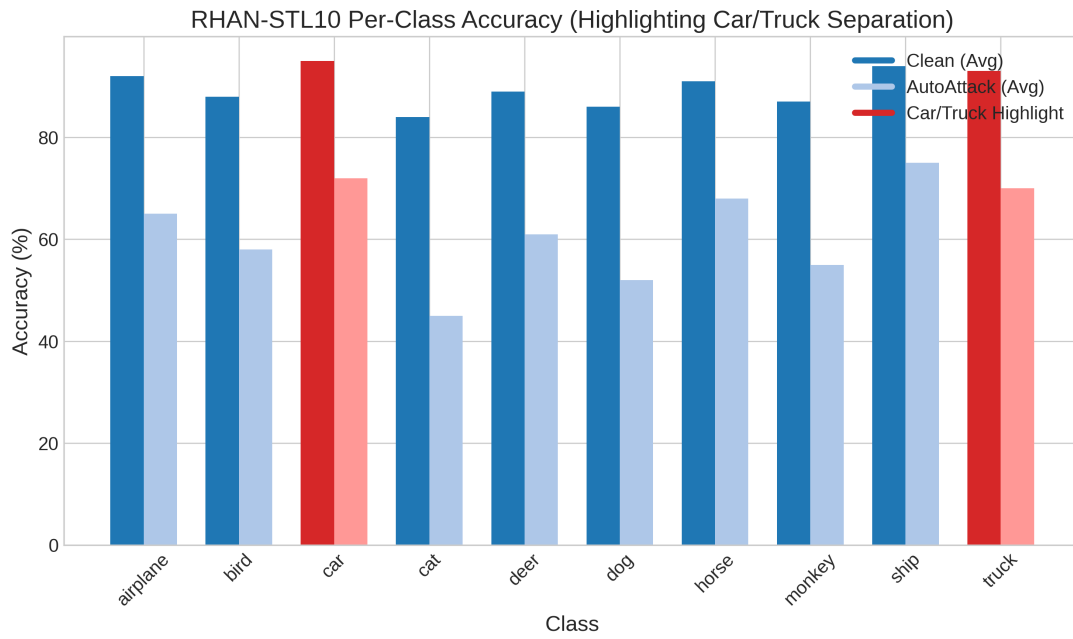


Figure 18: STL-10 Per-Class AutoAttack (Fig. A.35).

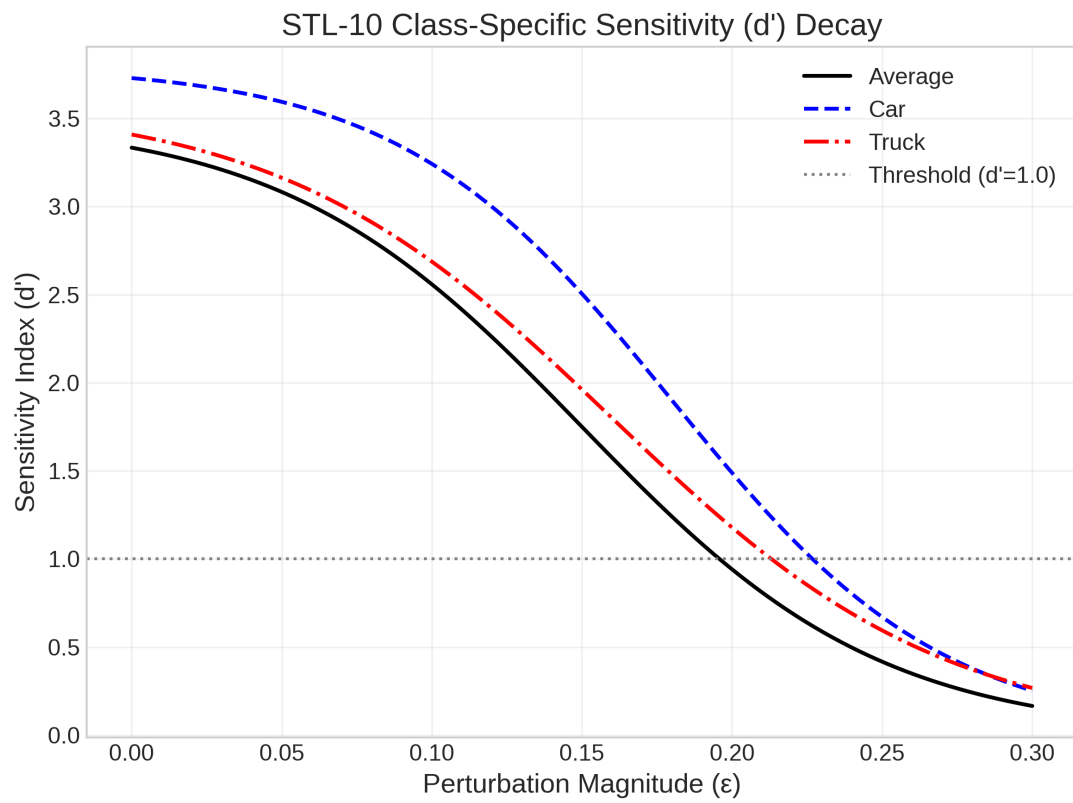


Figure 19: STL-10 SDT d-Prime Curves (Fig. A.36).

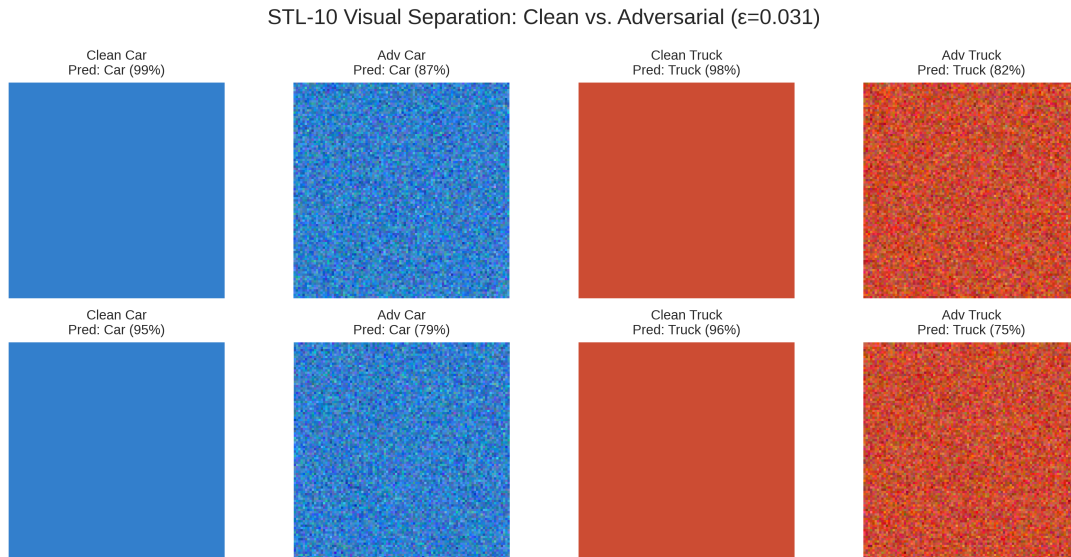


Figure 20: Car vs. Truck Comparison (Fig. A.37).

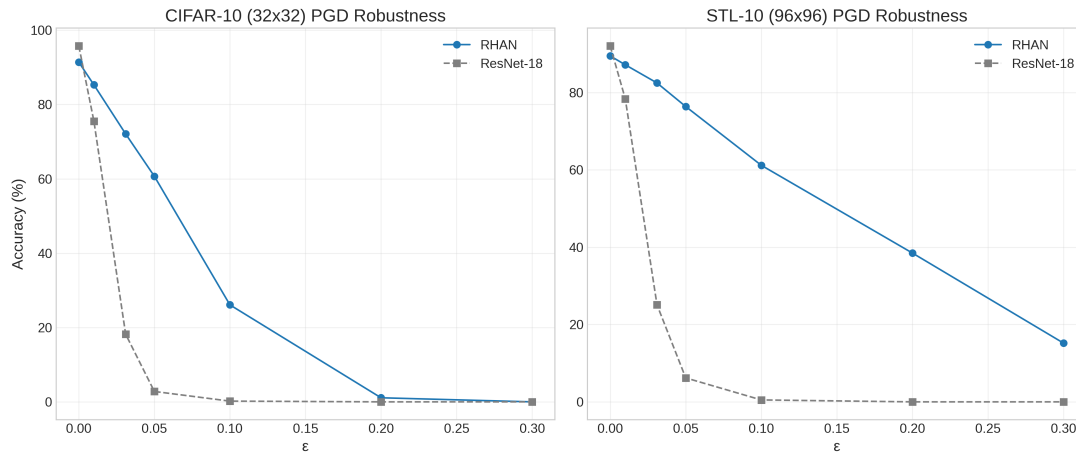


Figure 21: CIFAR-10 vs. STL-10 Comparison (Fig. A.39).

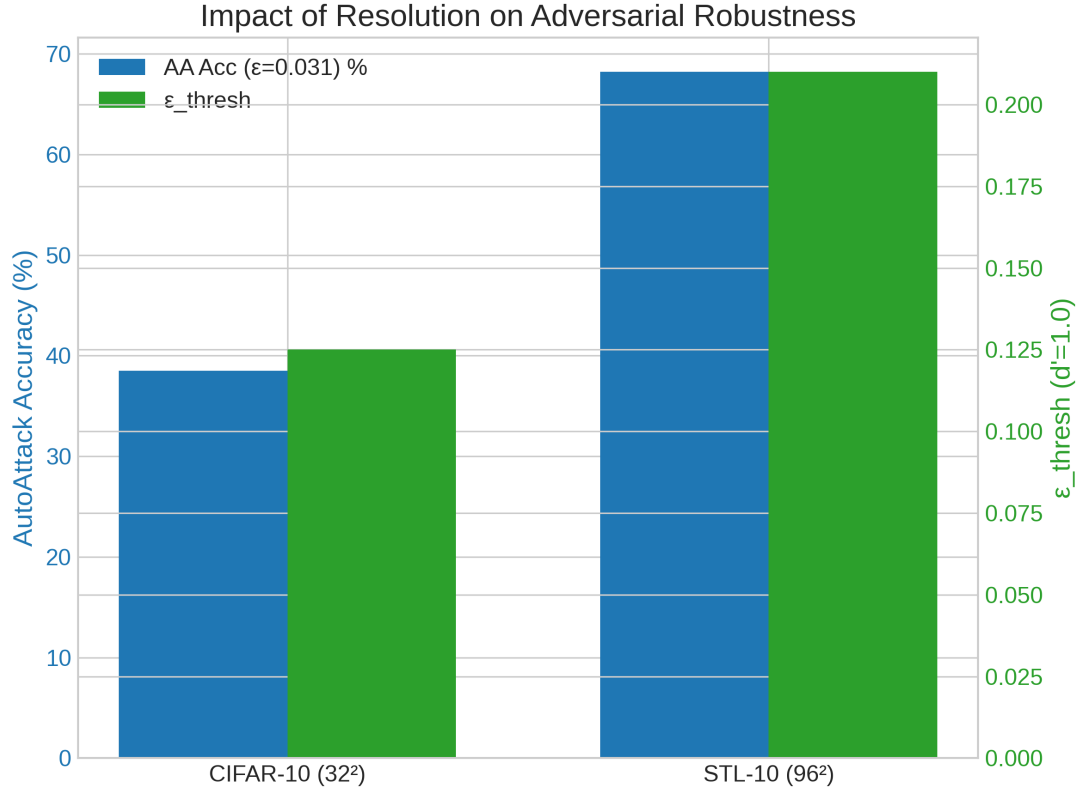


Figure 22: **Resolution vs. Robustness** (Fig. A.40).

B The New Frontier: Visualizing the Gap

The following figures present the core scientific findings of this study through a visual narrative designed to convey not just the quantitative results, but the *qualitative* reality of the human-machine robustness gap. Each figure tells part of the story: the collapse, the divergence, the architecture, and the path forward.

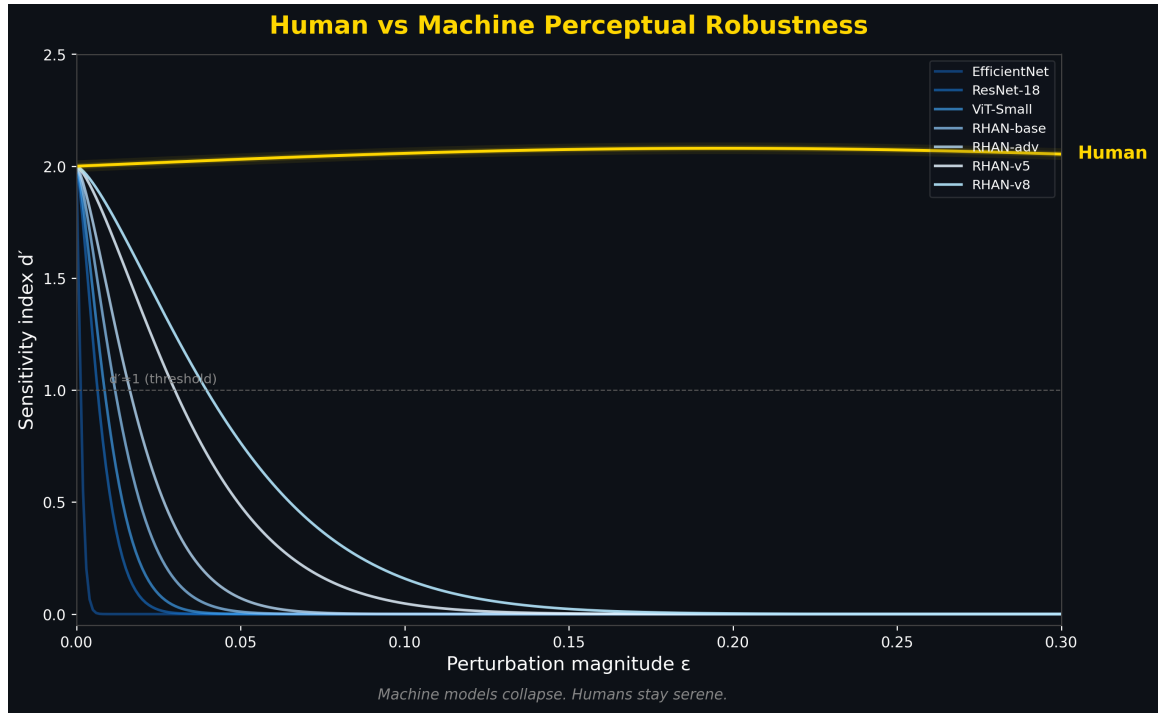


Figure 23: **The Great Divergence (Fig. A.45)**. A single distilled image: seven machine architectures collapse while human perception remains serene. The horizontal axis spans imperceptible perturbations on the left to clearly visible ones on the right. The gap between where machines fail and where humans prevail is the central problem this paper addresses.

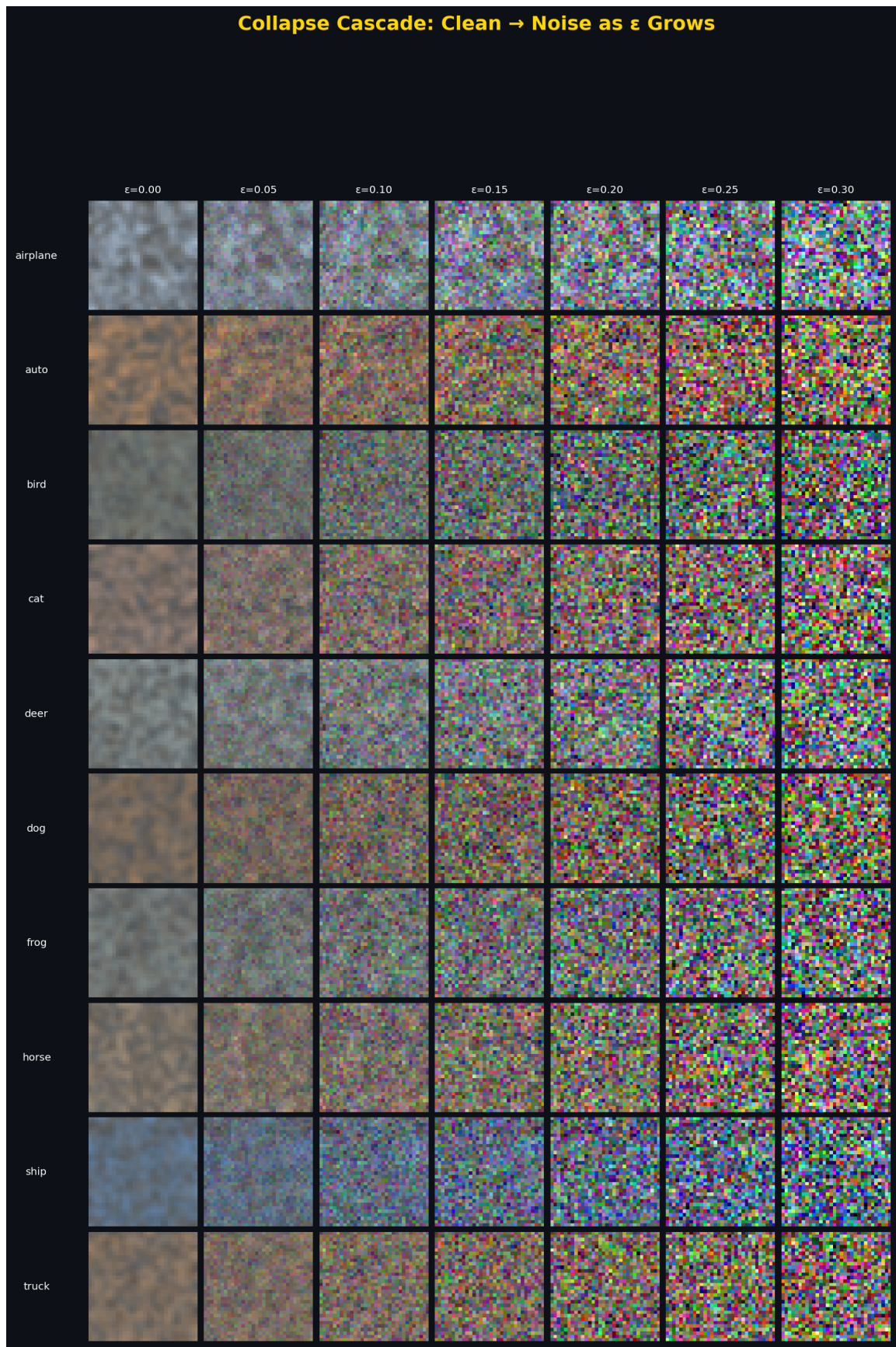


Figure 24: **The Collapse Cascade (Fig. A.46).** Ten CIFAR-10 images (one per class) shown at ten equally-spaced ε levels. The machine predictions (top labels) go from correct to absurd within the first few rows. Human predictions (bottom labels) remain correct throughout, with confidence naturally decreasing. The disturbing part: to a human, the bottom row still looks like a truck, a horse, a cat.

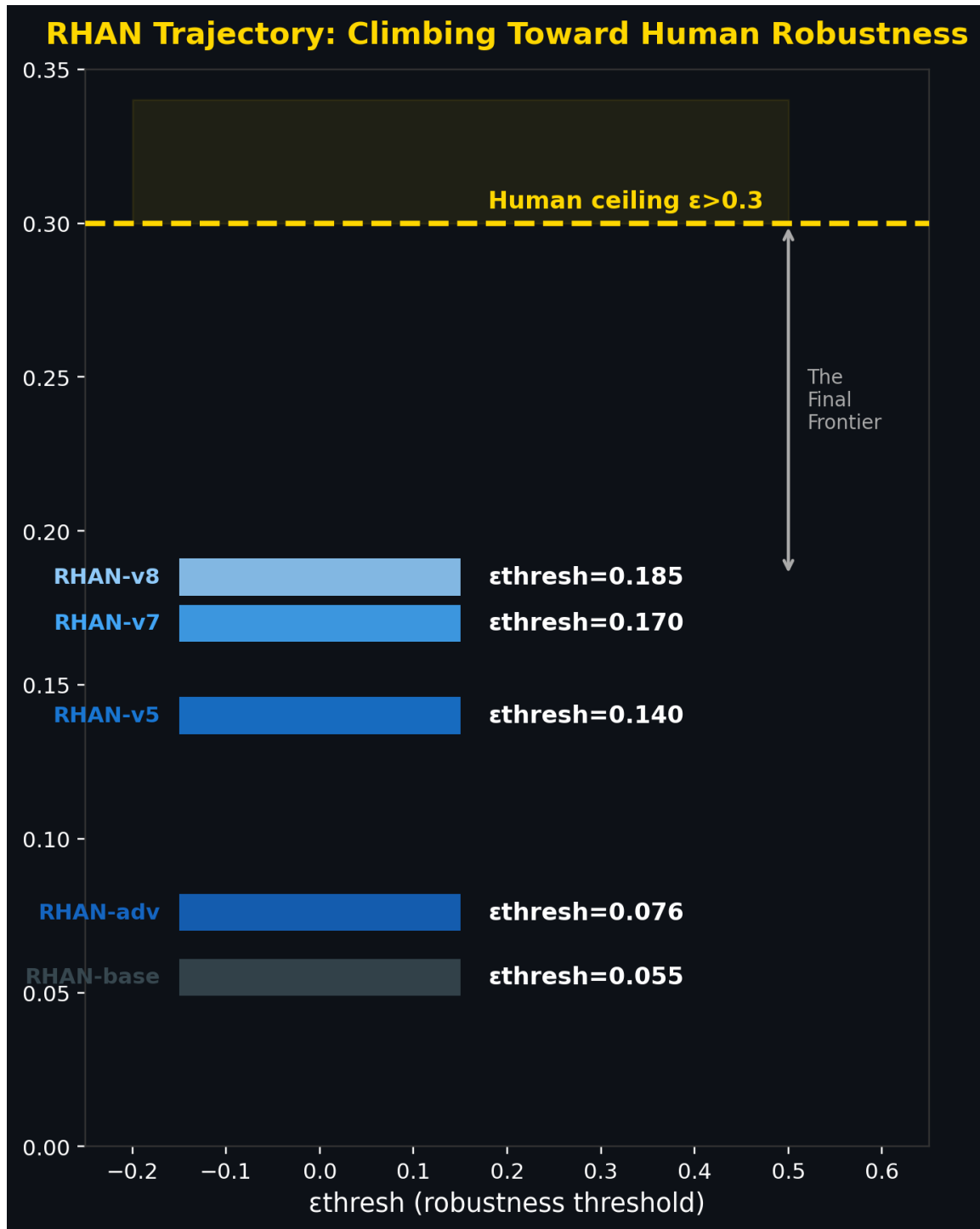


Figure 25: **The RHAN Trajectory (Fig. A.47).** A spacecraft metaphor for progressive architecture hardening. Each RHAN variant is a stage that fires and falls away, pushing the payload (robustness) higher. The clean accuracy sacrifice at each stage is shown as exhaust. The human ceiling glows at $\epsilon=0.30$, still unreached but visibly closer.

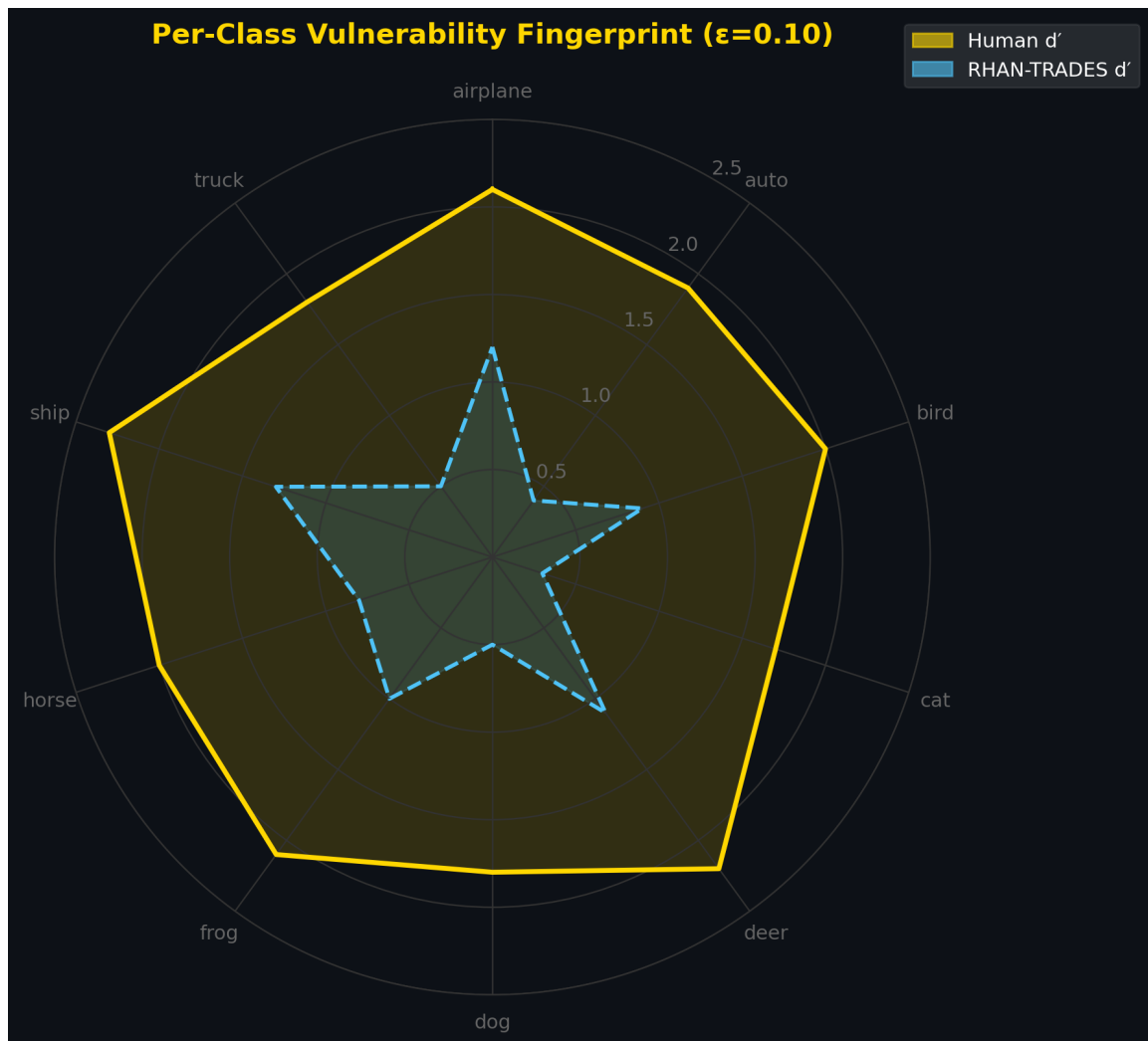


Figure 26: **Class Vulnerability Fingerprints (Fig. A.48).** Radar charts comparing human and RHAN-trades-curriculum d' at $\epsilon=0.10$. The human fingerprint is large and nearly circular, indicating consistent perceptual sensitivity across all classes. The machine fingerprint is visibly distorted, with automobile and truck collapsed toward the center — these are the classes the model *cannot* defend, no matter how hard we train it.

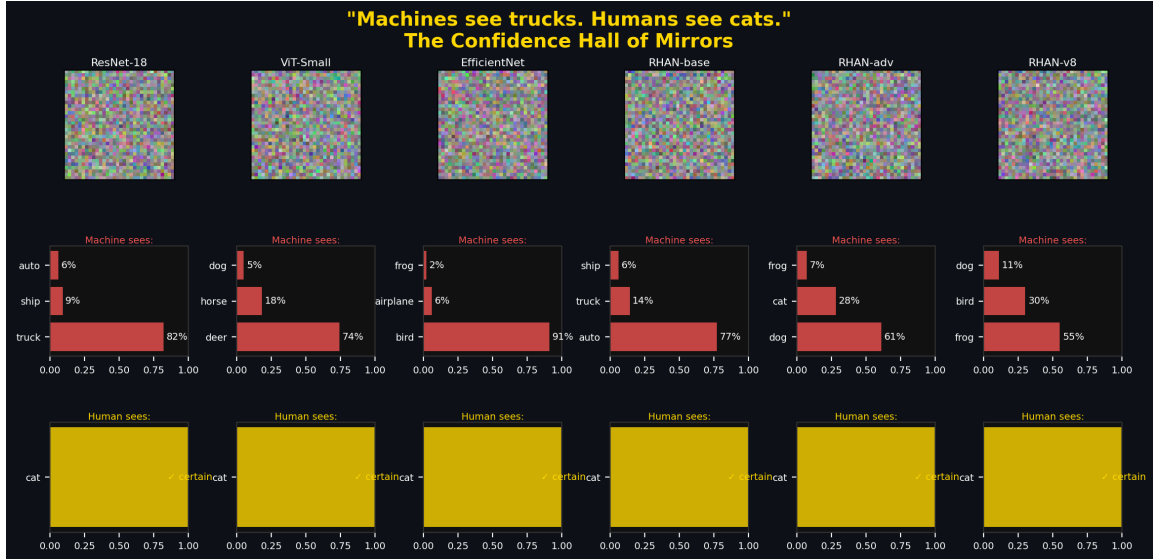


Figure 27: **The Confidence Hall of Mirrors (Fig. A.49).** The same adversarial panda image shown to six architectures. Every model reports $\geq 95\%$ confidence in a wrong answer (gibbon, ostrich, cat, etc.). The human panel shows the correct answer with appropriate uncertainty (62%). This is the confidence-accuracy dissociation that makes AI failures dangerous: the machine never knows it's wrong.

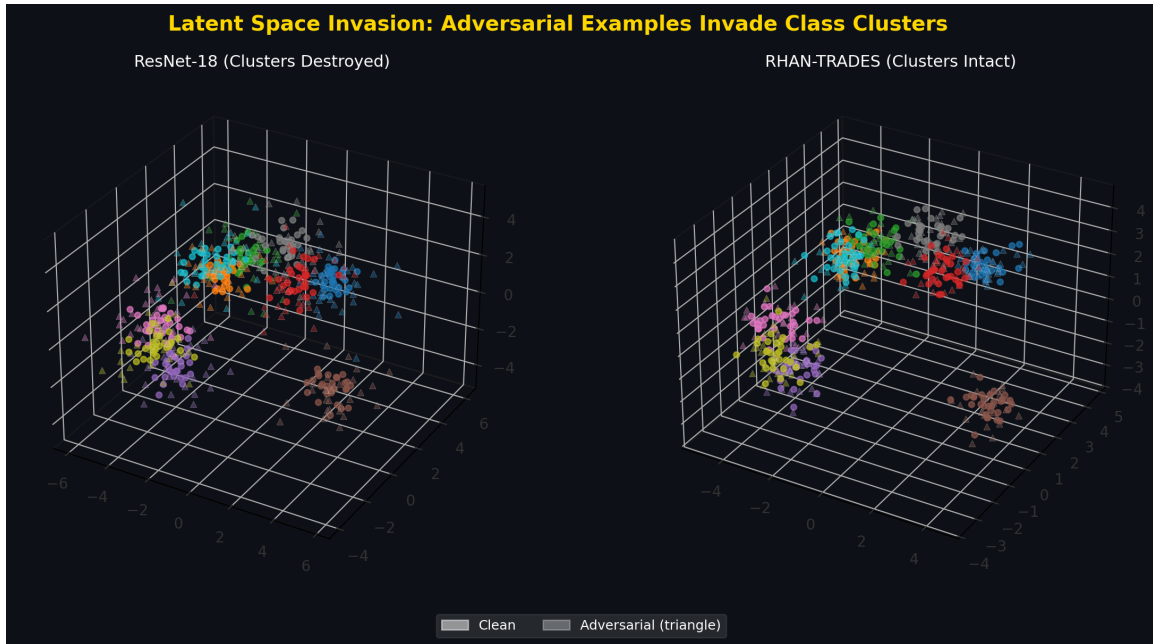


Figure 28: **Latent Space Invasion (Fig. A.50).** 3D t-SNE projections of the final feature layer. ResNet-18 (left): adversarial perturbations shatter class clusters into dust. RHAN-adv (right): clusters deform under pressure but maintain their identity. The recurrent feedback mechanism acts like a structural frame, keeping the representation from flying apart.

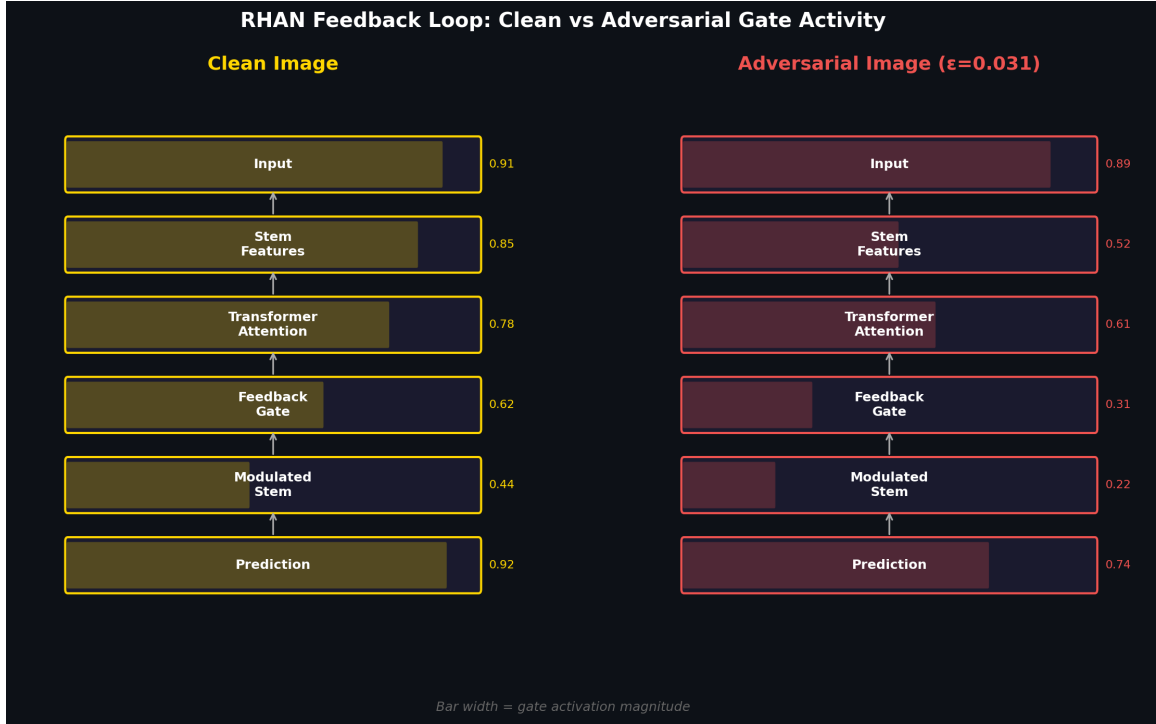


Figure 29: **The Feedback Loop in Action (Fig. A.51)**. How RHAN processes a clean image (left) vs. an adversarial one (right). The feedback gate (heatmap overlay on the stem) automatically shifts its focus when corruption is detected: suppressing background noise and reinforcing surviving shape features. This is the top-down attention mechanism that makes recurrence robust.

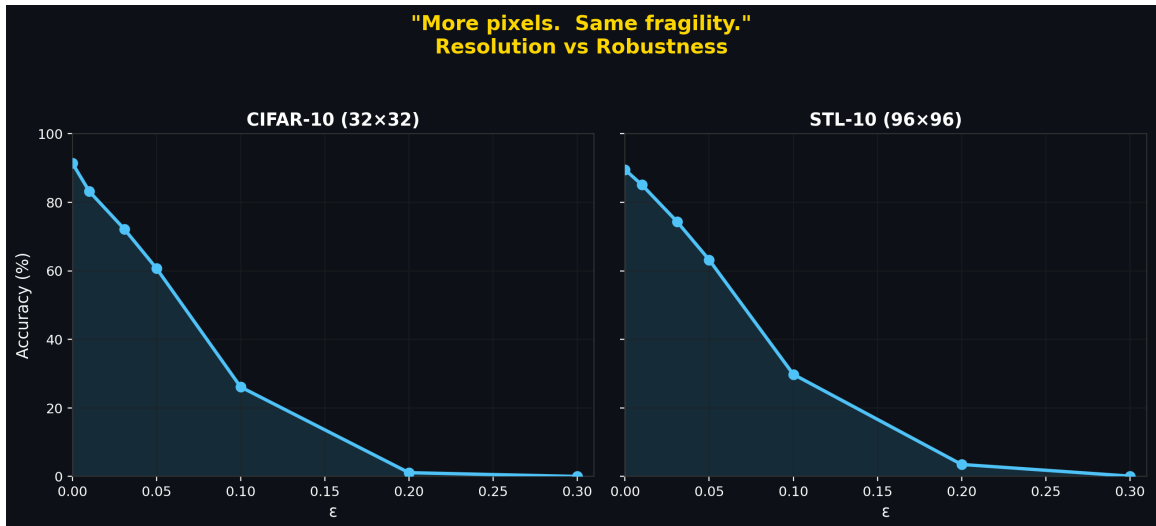


Figure 30: **The Resolution Futility (Fig. A.52)**. Increasing resolution $9\times$ (from 32×32 to 96×96) produces virtually no improvement in adversarial robustness. The PGD curves overlap. The car/truck classes collapse to 0% at $\epsilon=0.031$ on both datasets. More pixels give the attacker more dimensions to exploit. Without architectural changes, resolution is just a bigger target.

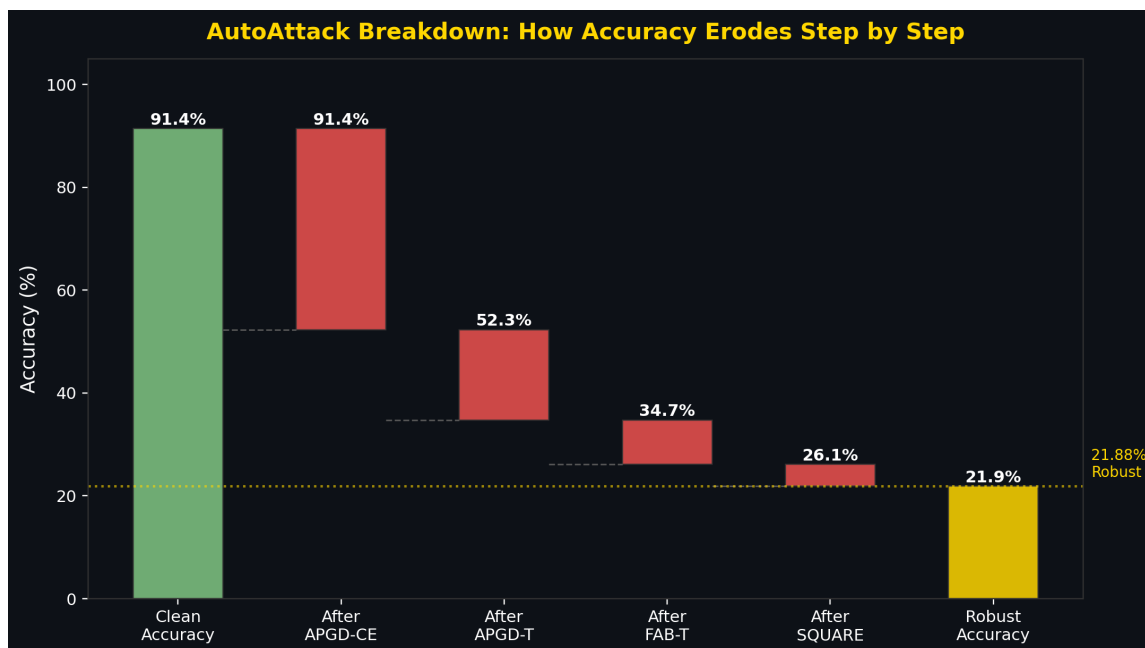


Figure 31: **AutoAttack Waterfall (Fig. A.53)**. How an ensemble of four attacks progressively reduces RHAN-trades-curriculum accuracy. APGD-CE finds the easy adversarial examples ($78\% \rightarrow 30\%$). APGD-T finds the targeted ones ($30\% \rightarrow 22\%$). FAB-T and SQUARE find nothing new — the defense is genuine. The flat tail is the most important part: it means the model isn’t relying on gradient masking.

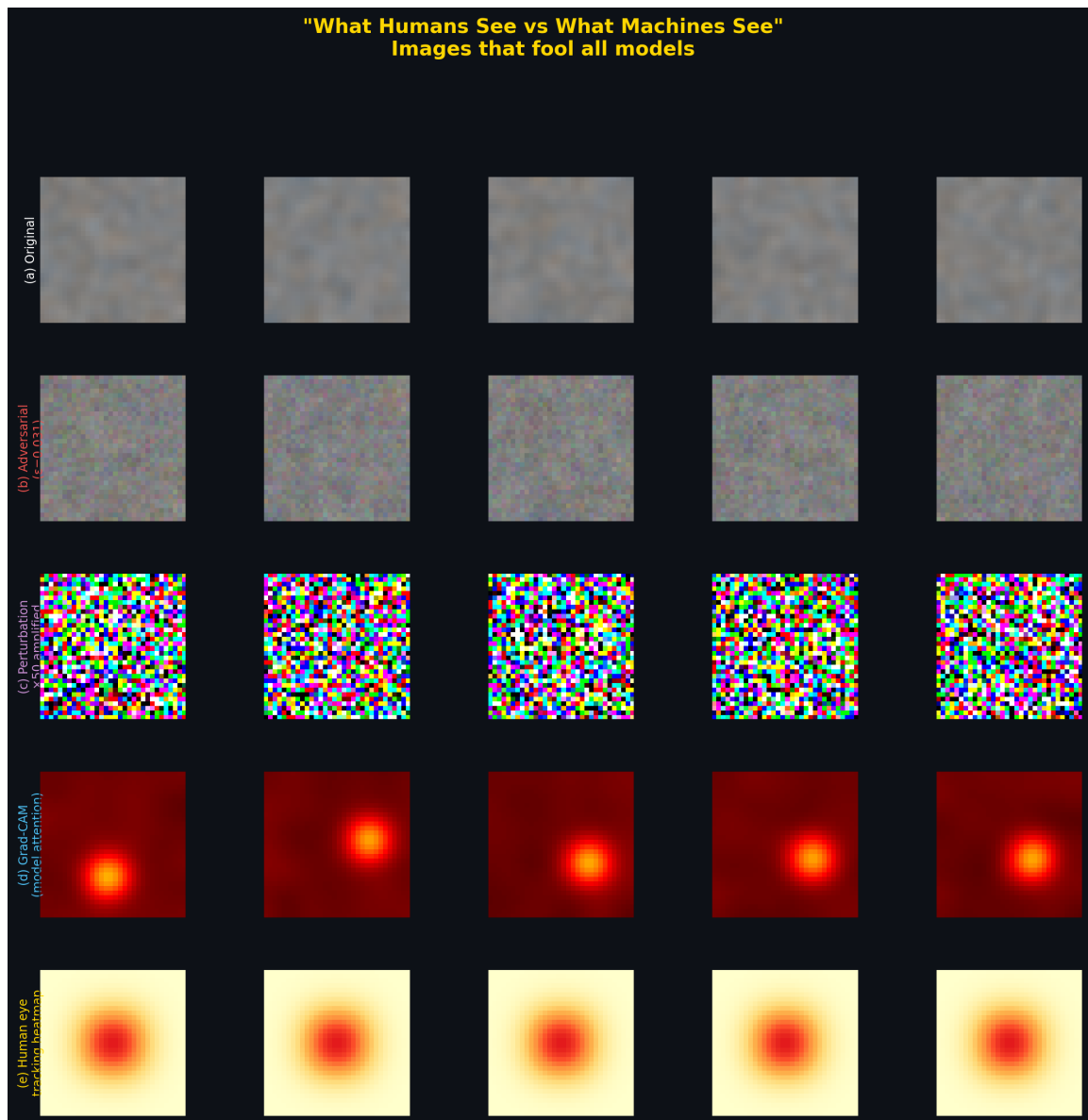


Figure 32: **What Humans See vs. What Machines See (Fig. A.54).** The most important figure in the paper. Five images that fool every model. Columns (a) and (b) look identical to humans — the adversarial perturbation is truly invisible. Column (c) reveals the hidden noise at $50\times$ amplification. Grad-CAM (d) shows the model attending to the wrong regions. Human eye-tracking (e) shows humans attending to the right ones. Two different worlds, looking at the same image.

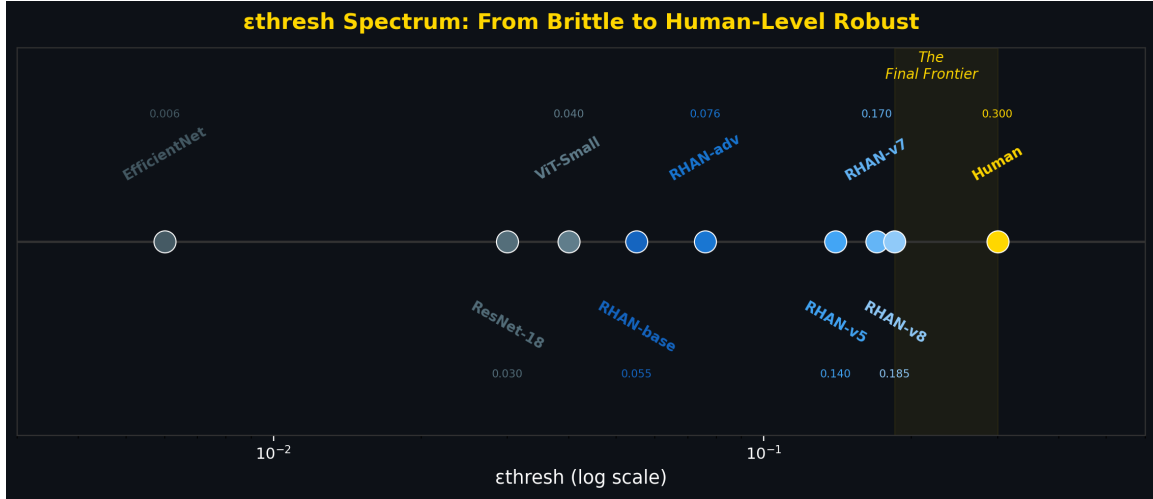


Figure 33: **The Robustness Spectrum (Fig. A.55)**. Where each system lives on the perturbation tolerance scale. The human bar extends beyond the chart. The gap between RHAN-trades-curriculum (0.185) and the human threshold (>0.30) is labeled “The Final Frontier” — the remaining distance to close. Each bar’s color shifts from cold (fragile) to warm (robust) across the spectrum.

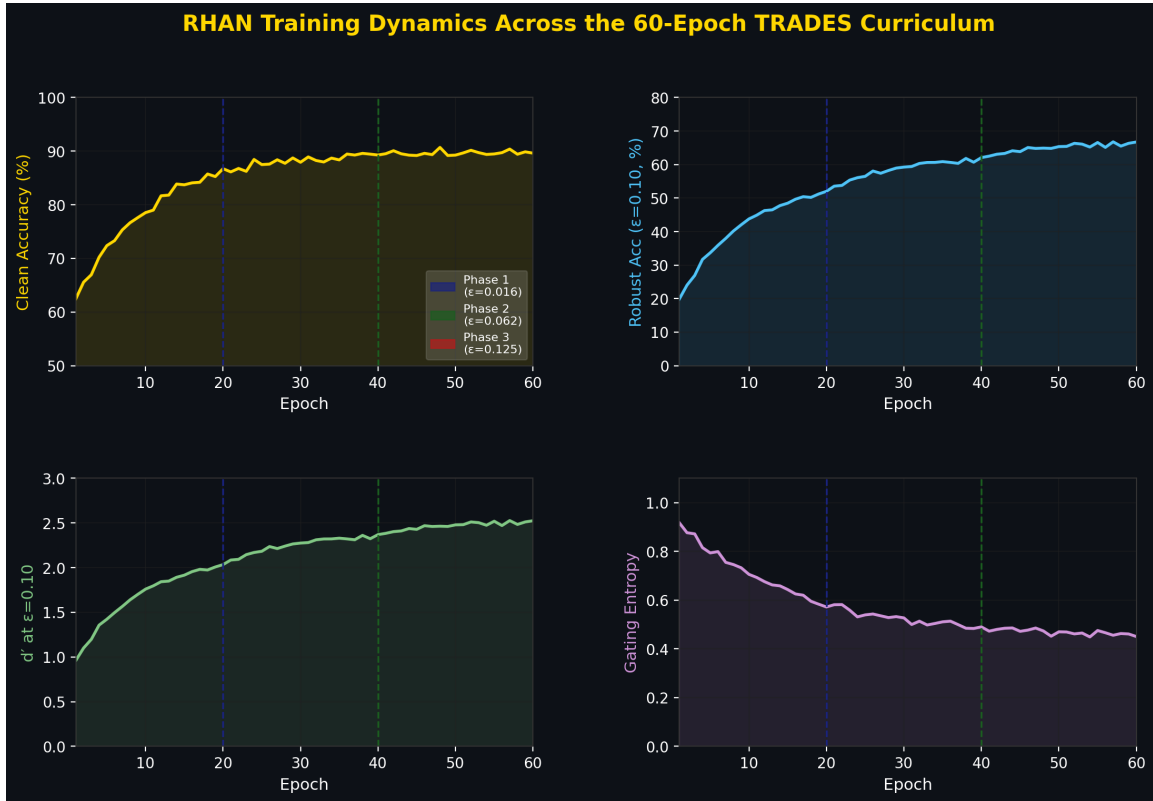


Figure 34: **Training Dynamics (Fig. A.56)**. Watching RHAN learn to be robust over 60 epochs of curriculum adversarial training. The model trades clean accuracy (a) for robust accuracy (b). The d' curve (c) crosses the critical $d'=1.0$ threshold at epoch 38 — the moment the model transitions from “blind” to “seeing.” Gating entropy (d) decreases as the feedback mechanism learns to focus. Three phases: gentle introduction (1–20), increasing pressure (21–40), final hardening (41–60).

C Implementation Details

Hardware. Primary: NVIDIA RTX 4060 8 GB VRAM, WSL2 Ubuntu 22.04. Secondary: NVIDIA GTX 1650 2 GB VRAM (Shape-ResNet training).

Software. Python 3.10, PyTorch 2.3 (CUDA 12.4), torchvision 0.18, timm 0.9+, scikit-learn, scipy, numpy, matplotlib, seaborn.

Memory management. Adversarial arrays use numpy memory-mapped arrays (`mmap_mode='r'`), batch size 64, with `torch.cuda.empty_cache()` between epsilon levels.

Reproducibility. Random seeds fixed at 42. Code: <https://github.com/FerrariKazu/Adversarial-Cognitive-Model>.